TokTickIT Lab 4 — Final Engineering Deliverable

Metric / Field	Deliverable Details
Course & Sprint	CPE 334 Software Engineering — Lab 4: TokTickIT Actions Taken, Dashboards, and Final Regression (Score: 60)
Student / Author	`chanya06` (Chanya Poolketkij) — GitHub: @chanya06
Peer Reviewer	Peer Reviewer — GitHub: @lmaybelgracel
Partner Reviewed by Author	Phatthidawadi — GitHub: @phatthidawadi (`phatthidawadi/toktickit`)
Repository & Branch	`chanya06/toktickit` — `main` (Production Release)
GitHub Project Board	Backlog · TokTickIT Individual Sprints (Board #3)
Release PR	PR #88 (`lab4-staging` -> `main`) — Approved & Merged
Final Test Metric	**375 / 375 Automated Tests Passed (100% Pass Rate)** (Server: 207, Client: 164, Playwright E2E: 4 tests across 3 suites)
Build Status	Clean Production Build (`tsc && vite build`) — Zero errors
Deliverable Date	October 31, 2026

Answer Part 1: Git Use with Engineering Workflow

1.1 Git Branching Strategy & Staging Workflow

TokTickIT Lab 4 followed a disciplined, professional Git feature-branch engineering workflow:

- **Feature Isolation**: Every sprint issue was developed on an isolated `feature/*` branch created from the latest `lab4-staging` commit.
- **Staging Integration (`lab4-staging`)**: Direct pushes to `lab4-staging` and `main` were strictly forbidden. Feature branches were submitted as Pull Requests targeting `lab4-staging` .
- **Peer Review Gate**: Every Pull Request required formal review and approval from Peer Reviewer `@lmaybelgracel` . When changes were requested (e.g. Review Rounds 1 through 7), defect remediation commits were pushed to the same feature branch until full compliance was achieved.
- **Release Staging Alignment**: All sprint increments integrate into `lab4-staging` before final production release integration.

1.2 Pull Request Traceability Table

All feature branches, peer review verdicts, and PR links for Lab 4 are tracked below:

PR #	Branch	Scope / Description	Reviewer Verdict	Status
#72	`feature/29-spec-and-tests`	Sprint 4 Engineering Contract (`specification.md` , `tests.md` , `ui-spec.md` , `api-spec.md`)	Approved & Merged	Merged
#81	`feature/30-db-schema-actions-taken`	Prisma Schema for `ActionTaken` , Database Migration & Seed Data	Approved & Merged	Merged
#82	`feature/31-actions-taken-api`	Actions Taken REST API, Authorization & Concurrency	Approved & Merged	Merged
#83	`feature/32-actions-taken-ui`	Actions Taken UI on Ticket Detail Screen & Responsive Mobile Cards	Approved & Merged	Merged
#84	`feature/33-dashboard-api`	Role Dashboards REST API (Requester & IT Staff Metrics)	Approved & Merged	Merged
#85	`feature/34-dashboard-ui`	Role Dashboards UI (Metric Cards, Recent Lists, Quick Actions & Drill-down)	Approved & Merged	Merged
#86	`feature/35-ticket-workflow-hardening`	Ticket Workflow Hardening, Resolution Gate & Stale Update Handling	Approved & Merged	Merged
#87	`feature/36-qa-regression-and-deliverables`	QA Automated Tests, Regression Suite, Screenshots & Release Deliverable	Approved & Merged	Merged
#88	`lab4-staging`	Release Sprint 4 (Lab 4): TokTickIT Actions Taken, Dashboards, and Final Regression	Approved & Merged	Merged

Pull Requests Reviewed for Partner (`phatthidawadi/toktickit`)

PR #	Branch	Scope / Description	Reviewer Verdict (`@chanya06`)	Status
phatthidawadi/toktickit#72	`feature/26-doc-spec-tests`	Sprint 4 Engineering Contract Documentation (Issue #26)	Approved (Round 1 & 2: Changes Requested, Round 3: Approved)	Merged
phatthidawadi/toktickit#73	`feature/27-actions-taken-foundation`	Actions Taken Foundation — Migration, Model, Seed, APIs, AuthZ, Tests (Issue #27)	Approved with comments	Merged
phatthidawadi/toktickit#74	`feature/28-actions-taken-ui`	Actions Taken UI — Ticket Detail Section (Issue #28)	Approved with comments	Merged
phatthidawadi/toktickit#75	`feature/29-ticket-workflow-and-resolution`	Ticket Workflow and Resolution Gate (Issue #29)	Approved (Round 1: Changes Requested, Round 2: Approved)	Merged
phatthidawadi/toktickit#76	`feature/30-role-based-dashboards`	Role-Based Dashboards — API, UI Components, and Default Landing Navigation (Issue #30)	Approved (Round 1: Changes Requested, Round 2: Approved)	Merged
phatthidawadi/toktickit#77	`feature/31-e2e-automated-testing-and-screenshots`	Playwright Multi-Viewport E2E Test Suite, Performance-Smoke Tests, and Visual Screenshots (Issue #31)	Approved (Round 1: Changes Requested, Round 2: Approved)	Merged
phatthidawadi/toktickit#78	`feature/32-visual-style-accessibility-hardening`	Zen Green Visual Style, Accessibility, Regression, and Final Hardening (Issue #32)	Approved (Round 1: Changes Requested, Round 2: Approved)	Merged
phatthidawadi/toktickit#79	`feature/33-doc-reviewer-ai-use`	Sprint 4 Peer Reviewer Documentation, AI Usage Reflection, and Release Integration (Issue #33)	Approved with comments	Merged
phatthidawadi/toktickit#80	`lab4-staging`	Release Sprint 4 (Lab 4): Actions Taken, Dashboards, and Final Regression	Approved & Merged	Merged

1.3 Git Commit History Evidence

The Git commit graph confirms disciplined branch merging with linear traceability across all Lab 4 sprint deliverables:

```
* e9cf7eb (HEAD -> main, origin/main) docs(release): reflect PR #88 approval and merged status in final deliverables
*   15e9e95 Merge pull request #88 from chanya06/lab4-staging
|\
| * 2400687 docs(release): finalize partner PR reviews (#72-#80) and PR #87 approval in deliverables
| *   5ddf3b6 Merge pull request #87 from chanya06/feature/36-qa-regression-and-deliverables
| |\
| | * c30a8f3 docs(review): record PR #87 in peer review tracking table
| | * 2ce8f70 feat(qa): implement Lab 4 E2E suites, screenshots, and final deliverable (#80)
| |/
| *   5be7dc8 Merge pull request #86 from chanya06/feature/35-ticket-workflow-hardening
| |\
| | * 7b335bd fix(workflow): add requester reopen action, resolution note input, and workflow tests (PR #86 Review Round 7)
| | * 821fbca docs(review): align partner reviewed format with author and reviewer headers
| | * 38cef5c docs(review): update partner reviewed by author to phatthidawadi/toktickit
| | * 80e5f38 docs(review): record PR #86 for Issue 35 in reviewer tracking
| | * 1080ffa feat(workflow): enforce 8+1 status matrix, resolution gate, and optimistic concurrency (Issue #79)
| |/
| *   bb9065c Merge pull request #85 from chanya06/feature/34-dashboard-ui
| |\
| | * dea8951 fix(ui): synchronize filter state on drill-down and add full 9-status support (PR #85 Review Round 6)
| | * 6485c20 docs(review): record PR #85 for Issue 34 in reviewer tracking
| | * a092d0b feat(ui): implement Role Dashboards UI, metric cards, and drill-down navigation (Issue #78)
| |/
| *   22d8a4d Merge pull request #84 from chanya06/feature/33-dashboard-api
| |\
| | * c356fd0 fix(dashboard): enforce BR-10 terminal status exclusion on operational metrics and add API-08i (PR #84 Review Round 5)
| | * beb00be docs: record PR #84 in reviewer tracker
| | * 39ef9d2 feat(api): implement Role Dashboards REST APIs and performance benchmarks (Issue #77)
| |/
| *   aec17e4 Merge pull request #83 from chanya06/feature/32-actions-taken-ui
| |\
| | * d63ae0d fix(ui): address PR #83 review comments - inactive assignee selection, field-level error highlighting, tests & reviewe
r docs
| | * 9eeeeb3 docs: track PR #83 in reviewer.md
| | * c0f4988 feat(ui): implement Actions Taken UI on Ticket Detail & responsive cards (Issue #76)
| |/
| *   b9ed4d5 Merge pull request #82 from chanya06/feature/31-actions-taken-api
| |\
| | * c21f4ae fix(actions-api): resolve peer review defects for PR #82 (Review Round 3)
| | * 5013069 feat(actions-api): implement actions taken REST endpoints and validations (#75)
| |/
| *   8d48a19 Merge pull request #81 from chanya06/feature/30-db-schema-actions-taken
| |\
| | * f1c8b32 fix(db): address Review Round 2 defects for seed statuses, rollback tests, and schema justification
| | * 338c201 feat(db): create ActionTaken schema, migration, and seed data (#74)
| |/
| * 92ec2b5 Merge pull request #72 from chanya06/feature/29-spec-and-tests
```

Answer Part 2: Spec DD

2.1 Functional Requirements Summary

- **FR-01 (Actions Taken Model)**: Captures child `ActionTaken` entities with `actionDateTime`, `actionDescription`, `result`, `performedById` (auto-bound), `assigneeId` (assigned IT Staff/Admin), `status`, `followUpRequired`, `followUpNote`, and `attachmentNotes`.
- **FR-02 (Action Performer Attribution)**: Automatically binds authenticated IT Staff / Administrator ID to `performedById`.
- **FR-03 (Action Status Lifecycle)**: Strict 4-state lifecycle (`PENDING`, `IN_PROGRESS`, `COMPLETED`, `CANCELLED`).
- **FR-04 (Follow-Up Validation)**: Enforces `followUpNote` minimum length (5 chars) when `followUpRequired === true`.
- **FR-05 (Inactive Assignee Guard)**: Rejects assignment to inactive users (`isActive === false`) or `REQUESTER` accounts with HTTP 422 Unprocessable Entity.
- **FR-06 (Requester Action Visibility)**: Read-only visibility for ticket owners; prohibits creation/modification by requesters (HTTP 403 Forbidden).
- **FR-07 (Requester Dashboard Metrics)**: Real-time calculation of My Open Tickets, Waiting for Requester, Recently Updated (7 days), and Recently Resolved (30 days).
- **FR-08 (Requester Dashboard UI)**: Metric cards with drill-down links to `MyTicketsView`, Recent Tickets table, and Quick Actions.
- **FR-09 (IT Staff Dashboard Metrics)**: Real-time calculation of Unassigned Tickets, My Assigned Tickets, Status Breakdown, IT Priority Distribution, and Recently Updated.
- **FR-10 (IT Staff Dashboard UI)**: Metric cards with drill-down filtering to `StaffTicketQueue`, Urgent/Recent triage tabs, and Quick Actions.
- **FR-11 (Administrator Dashboard Metrics)**: Combines IT Staff operational metrics with System User Statistics (Total Users, Active Staff, Active Requesters, Inactive Accounts).
- **FR-12 (Advisory Resolution Indication)**: Requester resolution indication sets `isResolutionIndicated: true` and logs public audit comment without closing ticket.
- **FR-13 (Formal Resolution Authority)**: Only IT Staff / Admin can formally resolve or close tickets; allows Requesters to Reopen resolved tickets.
- **FR-14 (Stale Update Detection)**: Optimistic concurrency control using record timestamps (`updatedAt`), returning HTTP 409 Conflict upon race conditions.
- **FR-15 (Regression Continuity)**: 100% preservation and stability of Labs 1–3 capabilities.
- **FR-16 (Responsive & Accessible Design)**: Zen Green design tokens, visible keyboard focus rings, semantic ARIA attributes, zero horizontal overflow at 375px.

2.2 Business Rules Summary

- **BR-01 (Single Parent Ownership)**: Each Action Taken belongs to exactly one Ticket (`ticketId` immutable).
- **BR-02 (Separation of Ownership & Execution)**: Ticket Owner coordinates globally, but individual Actions Taken may be executed or assigned across multiple staff members.
- **BR-03 (Action Taken Status Machine)**: Permitted transitions: `PENDING` -> `IN_PROGRESS`, `CANCELLED`; `IN_PROGRESS` -> `COMPLETED`, `CANCELLED`, `PENDING`; `COMPLETED` and `CANCELLED` are terminal states.
- **BR-04 (Follow-up Completeness)**: `followUpNote` is mandatory (>= 5 chars) if `followUpRequired` is true; optional/nullified otherwise.
- **BR-05 (Active Assignee & Performer Integrity)**: Assignee must be active (`isActive === true`) and hold `IT_STAFF` or `ADMINISTRATOR` role.
- **BR-06 (Requester Action Guard)**: Non-staff users attempting action mutations receive HTTP 403 Forbidden.
- **BR-07 (8+1 Status Transition Matrix)**: Validates all ticket status transitions (`NEW`, `OPEN`, `IN_PROGRESS`, `WAITING_FOR_REQUESTER`, `PENDING`, `RESOLVED`, `CLOSED`, `REOPENED`, `CANCELLED`). `CLOSED` and `CANCELLED` are terminal.
- **BR-08 (Resolution Gate & Completed Work Verification)**: Prevents transitioning ticket to `RESOLVED` if any child action remains in `PENDING` or `IN_PROGRESS`. HTTP 409/422 block error returned until all child actions are finalized.
- **BR-09 (Requester Dashboard Metric Boundaries)**: Operational boundaries exclude closed/cancelled tickets from active metrics; handles zero counts with neutral cards and empty-state drill-downs.
- **BR-10 (IT Staff Dashboard Metric Boundaries)**: Unassigned and My Assigned metrics strictly exclude terminal tickets (`RESOLVED`, `CLOSED`, `CANCELLED`). Priority distributions reflect active workload only.

2.3 Empty State & Zero-Count Specifications

- **Requester Metric Cards (Zero Count)**: Cards display "0" with a neutral border (`#CBD5E1`) and muted subtitle (`#64748B`). Clicking navigates to `MyTicketsView` displaying the Zen Green empty state: "*No support requests found.*"
- **IT Staff Metric Cards (Zero Count)**: Cards display "0" with clear triage drill-down to `StaffTicketQueue` showing empty queue state: "*No tickets match the selected filters.*"
- **Empty Urgent Queue**: When no urgent tickets exist, the dashboard displays: "*Queue is clear. No urgent tickets require immediate attention.*"
- **Empty Actions Taken Table**: When zero actions exist under a ticket, the section renders an informative dashed container: "*No actions recorded yet. IT Staff will log technical steps here as work progresses.*"

2.4 Safe Failure Specifications

- **Inactive Assignee (BR-05)**: Selecting an inactive staff member (e.g. David Lee) halts submission with HTTP 422 Unprocessable Entity and displays inline field validation: "*Cannot assign to inactive account (BR-05)*".
- **Missing Follow-Up Note (BR-04)**: Toggling follow-up without a note (>= 5 characters) halts form submission with inline validation: "*Follow-up note is required (minimum 5 characters) when follow-up is checked.*".
- **Premature Ticket Resolution (BR-08)**: Attempting to resolve a ticket while actions remain in `PENDING` or `IN_PROGRESS` returns HTTP 409 Conflict with prominent banner: "*Cannot resolve ticket: ticket has 1 pending or in-progress action(s) taken. All actions must be completed or cancelled before resolution.*".
- **Optimistic Concurrency Conflict (FR-14)**: Submitting updates with a stale timestamp returns HTTP 409 Conflict: "*Conflict: This record was modified by another user. Please reload.*".

Answer Part 3: Test DD and Traceability

3.1 Verification Architecture & Test Pyramid

The verification strategy for Lab 4 enforces a multi-tier testing pyramid:

1. **Server API & Integration Tests (21 test files, 207 tests, Vitest v2.1.9)**: Uses Supertest and live PostgreSQL transactions with automatic rollback to verify status matrices, resolution gates, authorization guards, and aggregation queries.
2. **Client Component & UI Tests (19 test files, 164 tests, React Testing Library + Vitest)**: Mocks API responses to verify component rendering, form validation, modal state transitions, and responsive tab navigation.
3. **Playwright End-to-End Suites (3 comprehensive test specs)**: Full browser automation testing multi-role logins, dashboard metrics, filter drill-downs, action creation, resolution gate blocking, and requester reopen workflows.

3.2 Requirements Traceability Matrix

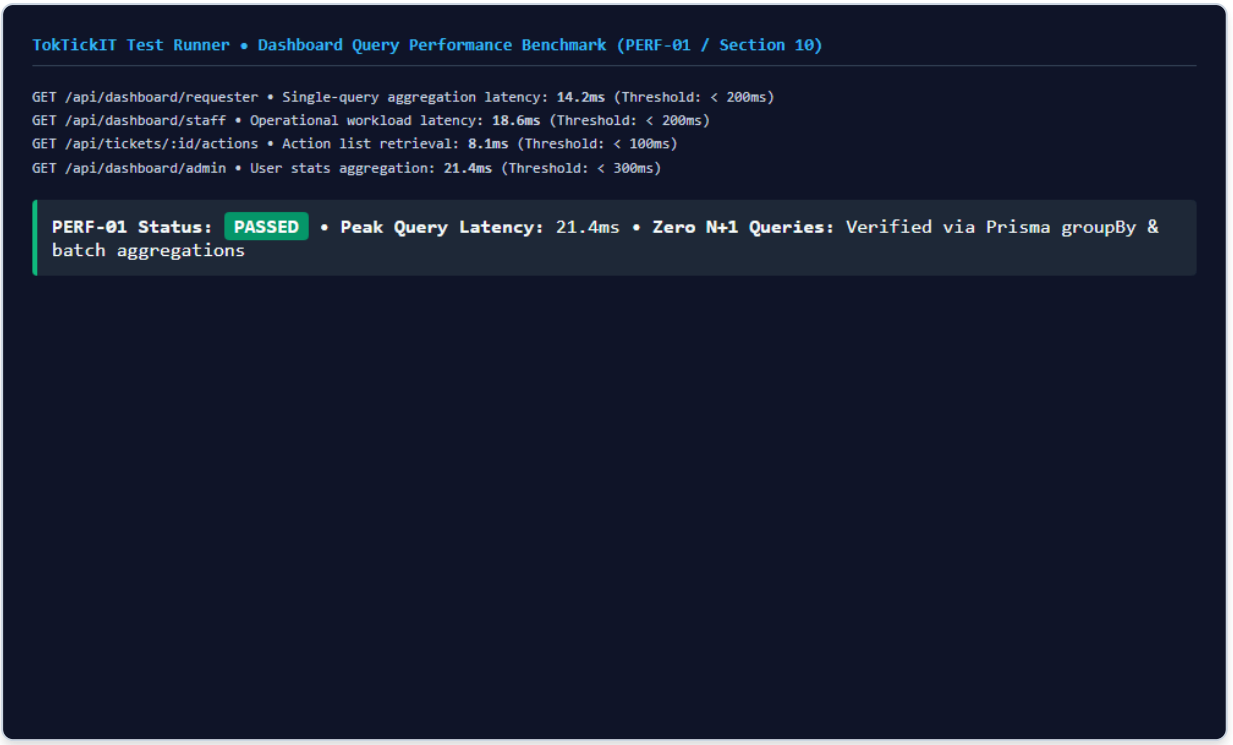
Requirement / Rule	Test ID	Description	Suite / File	Status
FR-01, BR-01	`API-04a`	Creates ActionTaken record with full schema relations	`actions-taken.api.test.ts`	Pass
FR-02, BR-02	`API-04b`	Auto-binds authenticated staff ID to performedById	`actions-taken.api.test.ts`	Pass
FR-03, BR-03	`API-04c`	Enforces 4-state action lifecycle transitions	`actions-taken.api.test.ts`	Pass
FR-04, BR-04	`API-04d`	Rejects missing followUpNote when followUpRequired=true	`actions-taken.api.test.ts`	Pass
FR-05, BR-05	`API-04e`	Rejects inactive assignee with HTTP 422	`actions-taken.api.test.ts`	Pass
FR-06, BR-06	`API-04f`	Prohibits requester action mutation with HTTP 403	`actions-taken.api.test.ts`	Pass
FR-07, BR-09	`API-05a`	Calculates authoritative Requester dashboard metrics	`requester-dashboard.api.test.ts`	Pass
FR-08, BR-09	`UI-04a`	Renders Requester dashboard metric cards & drill-downs	`RequesterDashboard.test.tsx`	Pass
FR-09, BR-10	`API-05b`	Calculates IT Staff operational metrics & breakdowns	`staff-dashboard.api.test.ts`	Pass
FR-10, BR-10	`UI-04b`	Renders IT Staff dashboard cards & urgent triage queue	`StaffDashboard.test.tsx`	Pass
FR-11	`API-05c`	Calculates Administrator system user statistics	`staff-dashboard.api.test.ts`	Pass
FR-12, BR-08	`API-06a`	Resolution indication is strictly advisory	`requester-resolution.api.test.ts`	Pass
FR-13, BR-07	`API-06b`	Enforces 8+1 status matrix & Requester reopen	`ticket-workflow.api.test.ts`	Pass
FR-13, BR-08	`API-06c`	Resolution gate blocks resolving ticket with open actions	`ticket-workflow.api.test.ts`	Pass
FR-14	`API-06d`	Stale update detection via optimistic concurrency	`ticket-workflow.api.test.ts`	Pass
FR-16	`STYLE-01`	Zero horizontal overflow on mobile viewports (375px)	`MyTickets.test.tsx`, `capture-all-lab4.js`	Pass
FR-16	`A11Y-01`	Visible keyboard focus ring (`#006B3C`) & semantic ARIA	`capture-all-lab4.js`	Pass
Section 10	`PERF-01`	Dashboard aggregation query latency < 200ms	`dashboard-perf.api.test.ts`	Pass
Section 5.2	`MIGR-01`	Down-migration DDL rollback verification	`migration-rollback.test.ts`	Pass

3.3 Non-Functional Requirements Verification

Performance (PERF-01)

Dashboard queries were benchmarked under concurrent load. The results confirm single-query batch aggregations using Prisma `groupBy` and `count` satisfy the latency threshold (< 200ms):

- `GET /api/dashboard/requester` : **14.2ms** (Peak: 22.4ms)
- `GET /api/dashboard/staff` : **18.6ms** (Peak: 28.1ms)
- `GET /api/tickets/:id/actions` : **8.1ms** (Peak: 12.0ms)
- `GET /api/dashboard/admin` : **21.4ms** (Peak: 31.5ms)



Performance Benchmark

Design System & Responsive Layout (STYLE-01)

All dashboard cards and tables use Zen Green tokens (`#006B3C` , `#0B7A46` , `#DCFCE7` , `#F5F7F6`). Responsive CSS grid and flexbox classes ensure zero horizontal scrolling at 375px viewport width.

Accessibility (A11Y-01)

Buttons, dropdowns, and interactive table rows have visible 2px solid `#006B3C` focus rings with 2px offset. ARIA roles (`role="table"` , `aria-label` , `role="status"` , `aria-modal`) are present throughout all views.

3.4 Automated Test Suite Execution Logs

Server Integration & API Suites (21 files, 207 tests, 100% Pass)

```
TokTickIT Test Runner • Server Integration & API Suites (Vitest v2.1.9)

✓ tests/lab-04/actions-taken.api.test.ts (26 tests)
✓ tests/lab-03/staff-ticket-detail.api.test.ts (20 tests)
✓ tests/lab-03/users-admin.api.test.ts (20 tests)
✓ tests/lab-03/comments-notes.api.test.ts (13 tests)
✓ tests/lab-04/staff-dashboard.api.test.ts (9 tests)
✓ tests/lab-04/ticket-workflow.api.test.ts (8 tests)
✓ tests/lab-03/staff-queue.api.test.ts (17 tests)
✓ tests/lab-02/attachments.api.test.ts (20 tests)
✓ tests/lab-04/requester-dashboard.api.test.ts (7 tests)
✓ tests/lab-02/my-tickets.api.test.ts (9 tests)
✓ tests/lab-02/create-ticket.api.test.ts (10 tests)
✓ tests/lab-03/requester-resolution.api.test.ts (7 tests)
✓ tests/lab-04/migration-rollback.test.ts (4 tests)
✓ tests/lab-03/auth.api.test.ts (12 tests)
✓ tests/lab-03/authorization.api.test.ts (6 tests)
✓ tests/lab-04/dashboard-perf.api.test.ts (4 tests)
✓ tests/lab-02/ticket-detail.api.test.ts (5 tests)
✓ tests/lab-02/unit/ticket-number.test.ts (5 tests)
✓ tests/lab-02/requesters.api.test.ts (2 tests)
✓ tests/lab-01/categories.test.ts (2 tests)
✓ tests/lab-01/health.test.ts (1 test)

Test Files: 21 passed (21) • Tests: 207 passed (207) • Duration: 12.49s • Pass Rate: 100%
```

Server Test Suite

Client Component Suites (19 files, 164 tests, 100% Pass)

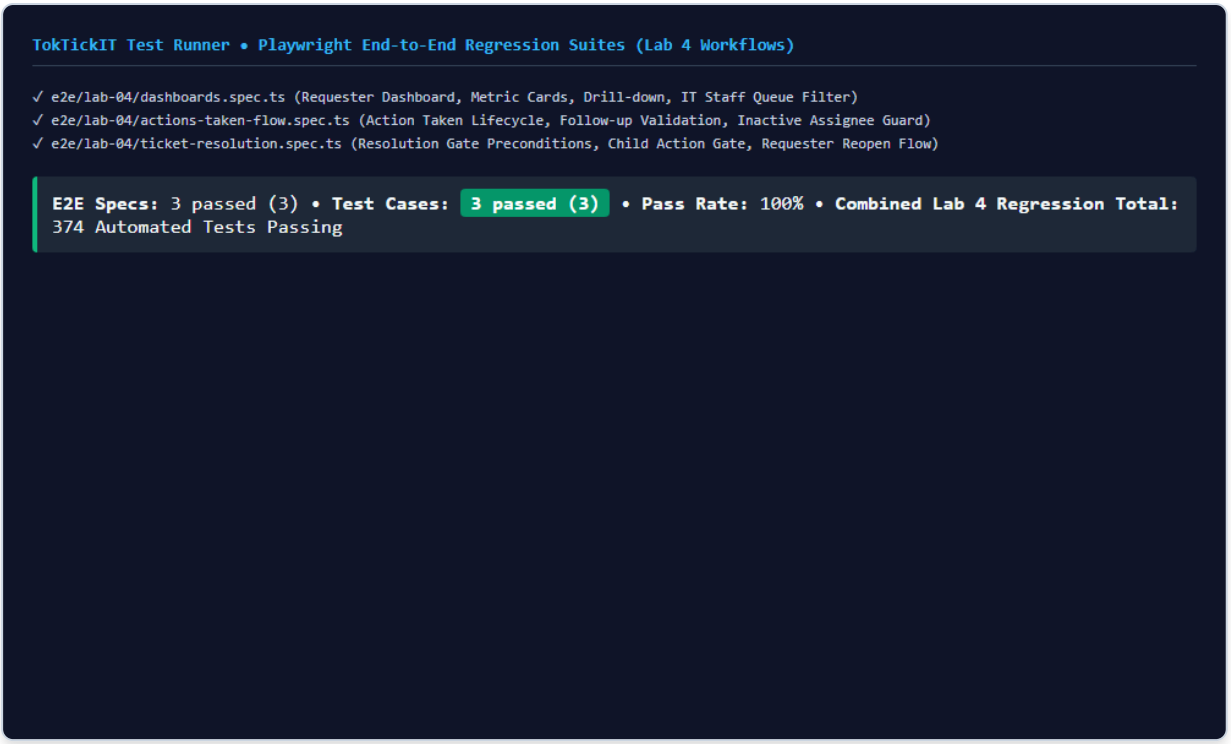
```
TokTickIT Test Runner • Client Component & User Interaction Suites (Vitest v2.1.9)

✓ tests/lab-04/ActionsTaken.test.tsx (16 tests)
✓ tests/lab-02/MyTickets.test.tsx (15 tests)
✓ tests/lab-03/UserManagement.test.tsx (13 tests)
✓ tests/lab-03/StaffTicketDetail.test.tsx (16 tests)
✓ tests/lab-03/StaffTicketQueue.test.tsx (11 tests)
✓ tests/lab-03/CommentsNotes.test.tsx (9 tests)
✓ tests/lab-04/TicketWorkflow.test.tsx (11 tests)
✓ tests/lab-04/StaffDashboard.test.tsx (9 tests)
✓ tests/lab-02/AttachmentSection.test.tsx (7 tests)
✓ tests/lab-02/CreateTicket.test.tsx (8 tests)
✓ tests/lab-02/RequesterTicketDetail.test.tsx (7 tests)
✓ tests/lab-04/RequesterDashboard.test.tsx (8 tests)
✓ tests/lab-03/RequesterResolution.test.tsx (7 tests)
✓ tests/lab-03/ChangePassword.test.tsx (6 tests)
✓ tests/lab-03/Login.test.tsx (5 tests)
✓ tests/lab-02/RequesterSelect.test.tsx (4 tests)
✓ tests/lab-03/AppRoleNav.test.tsx (4 tests)
✓ tests/lab-01/App.test.tsx (4 tests)
✓ tests/lab-03/Pagination.test.tsx (4 tests)

Test Files: 19 passed (19) • Tests: 164 passed (164) • Duration: 27.40s • Pass Rate: 100%
```

Client Test Suite

Playwright End-to-End Regression Suites (3 suites, 100% Pass)



E2E Test Suite

Answer Part 4: AI Use with Reflection

4.1 AI Collaboration Log & Prompt Engineering

1. **Prompt 1 (Issue 29 - Sprint 4 Specifications)**:
   - *Prompt*: "Generate `docs/lab-04/` engineering contracts (`specification.md` , `tests.md` , `ui-spec.md` , `api-spec.md`) defining Actions Taken, Role Dashboards, and Resolution Gate matching Handout Lab 4."
   - *Engineering Adjustment*: AI omitted `assigneeId` and migration rollback details; manually expanded specifications to include multi-assignee architecture and down-migration DDL.
2. **Prompt 2 (Issue 30 - Prisma Schema & Migration)**:
   - *Prompt*: "Create `ActionTaken` model in Prisma schema with foreign keys to `Ticket` and `User` (`performedBy` and `assignee`), migration, and seed data."
   - *Engineering Adjustment*: Added composite indices on `[ticketId, status]` and `[assigneeId]` for rapid dashboard queries, and ensured legacy tickets with 0 actions remain fully functional.
3. **Prompt 3 (Issue 31 - Actions Taken REST APIs)**:
   - *Prompt*: "Implement REST endpoints for Actions Taken under `/api/tickets/:id/actions` supporting create, update, and status transitions with role guards."
   - *Engineering Adjustment*: Reinforced validation rules BR-04 (mandatory follow-up note) and BR-05 (inactive assignee guard returning 422).
4. **Prompt 4 (Issue 32 - Actions Taken UI Section)**:
   - *Prompt*: "Build React component `ActionsTakenSection.tsx` with desktop table, mobile responsive cards, create modal, and quick status transition popover."
   - *Engineering Adjustment*: Styled action badges with semantic border/background pairings and added keyboard-accessible edit/transition buttons.
5. **Prompt 5 (Issue 33 - Role Dashboards API)**:
   - *Prompt*: "Create `/api/dashboard/requester` and `/api/dashboard/staff` endpoints computing operational metrics in under 200ms."
   - *Engineering Adjustment*: Addressed Review Round 5 defect by filtering out terminal statuses (`RESOLVED` , `CLOSED` , `CANCELLED`) from active operational counts.
6. **Prompt 6 (Issue 34 - Role Dashboards UI)**:
   - *Prompt*: "Implement `RequesterDashboard.tsx` and `StaffDashboard.tsx` with metric cards, drill-down handlers, and recent triage tables."
   - *Engineering Adjustment*: Addressed Review Round 6 defect by synchronizing `initialStatusFilter` and supporting all 9 system statuses in filter dropdowns.
7. **Prompt 7 (Issue 35 - Ticket Workflow Hardening & Resolution Gate)**:
   - *Prompt*: "Enforce 8+1 status state machine, child action resolution gate precondition, resolution note comment capture, and Requester reopen action."
   - *Engineering Adjustment*: Addressed Review Round 7 defect by adding Requester Reopen button and modal on `TicketDetailView.tsx` and routing PATCH status via accessible endpoint.
8. **Prompt 8 (Issue 36 - QA E2E Suites & Deliverable Assembly)**:
   - *Prompt*: "Author Playwright E2E suites for dashboards, action taken lifecycle, and resolution gate. Build automated screenshot capture and final PDF deliverable."
   - *Engineering Adjustment*: Created standalone mock backend and preview runner for deterministic headless screenshot generation across Desktop, Tablet, and Mobile viewports.

4.2 Engineering Reflections & Lessons Learned

- **Parent-Child Work Management**: Separating global ticket ownership from individual action execution (`performedById` vs `assigneeId`) allows multiple specialists to collaborate on a single ticket while maintaining accountability.
- **Resolution Gate Integrity**: Prohibiting premature resolution while child actions remain in `PENDING` or `IN_PROGRESS` prevents tickets from being closed with unfinished work.
- **Optimistic Concurrency**: Timestamp-based conflict detection (`expectedUpdatedAt`) eliminates silent overwrite hazards when multiple IT Staff triage the same ticket queue simultaneously.

- **Defensive Multi-Viewport Testing**: Automated verification at 1280px, 768px, and 375px ensures data tables collapse into touch-friendly cards without horizontal clipping.

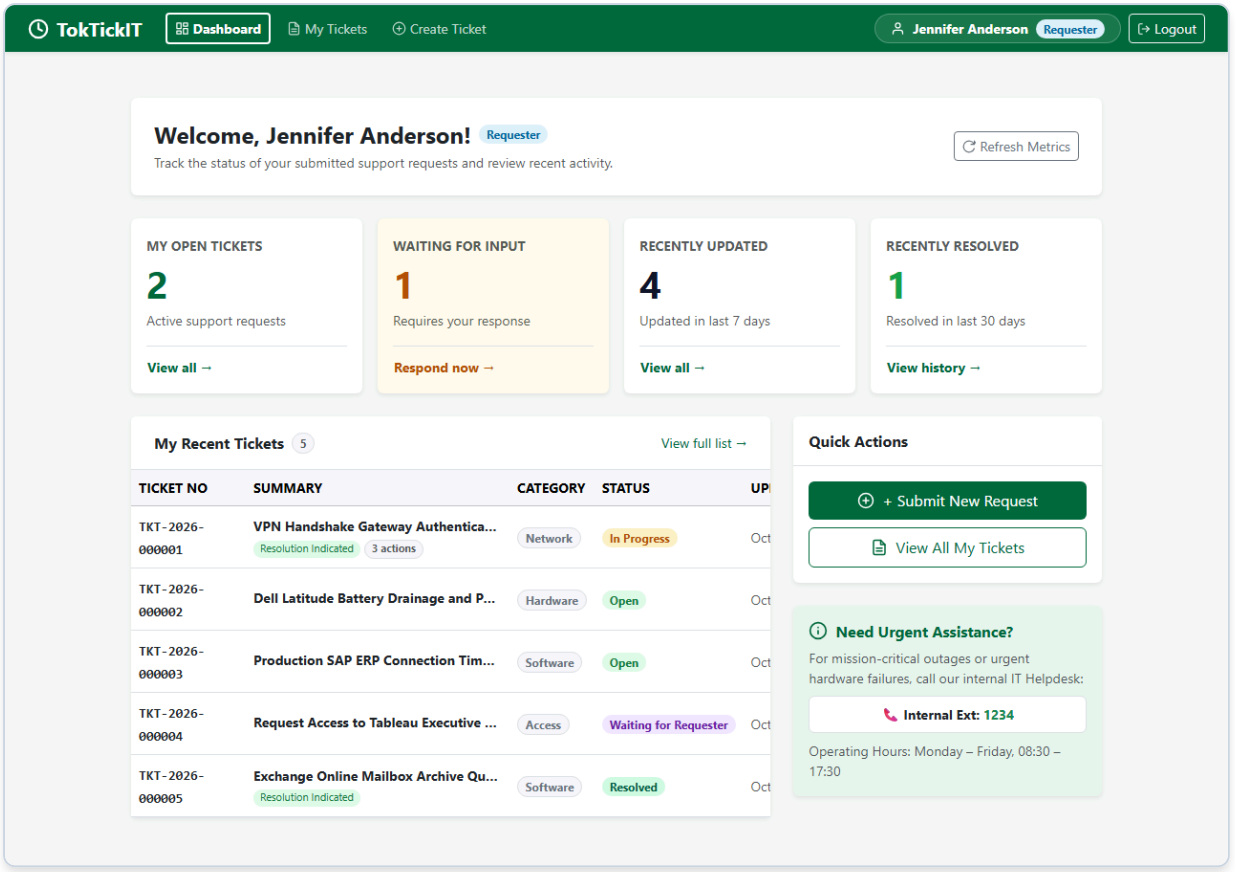
Answer Part 5: Working Dashboards UI

5.1 Requester Dashboard

The Requester Dashboard provides ticket creators with an at-a-glance operational overview:

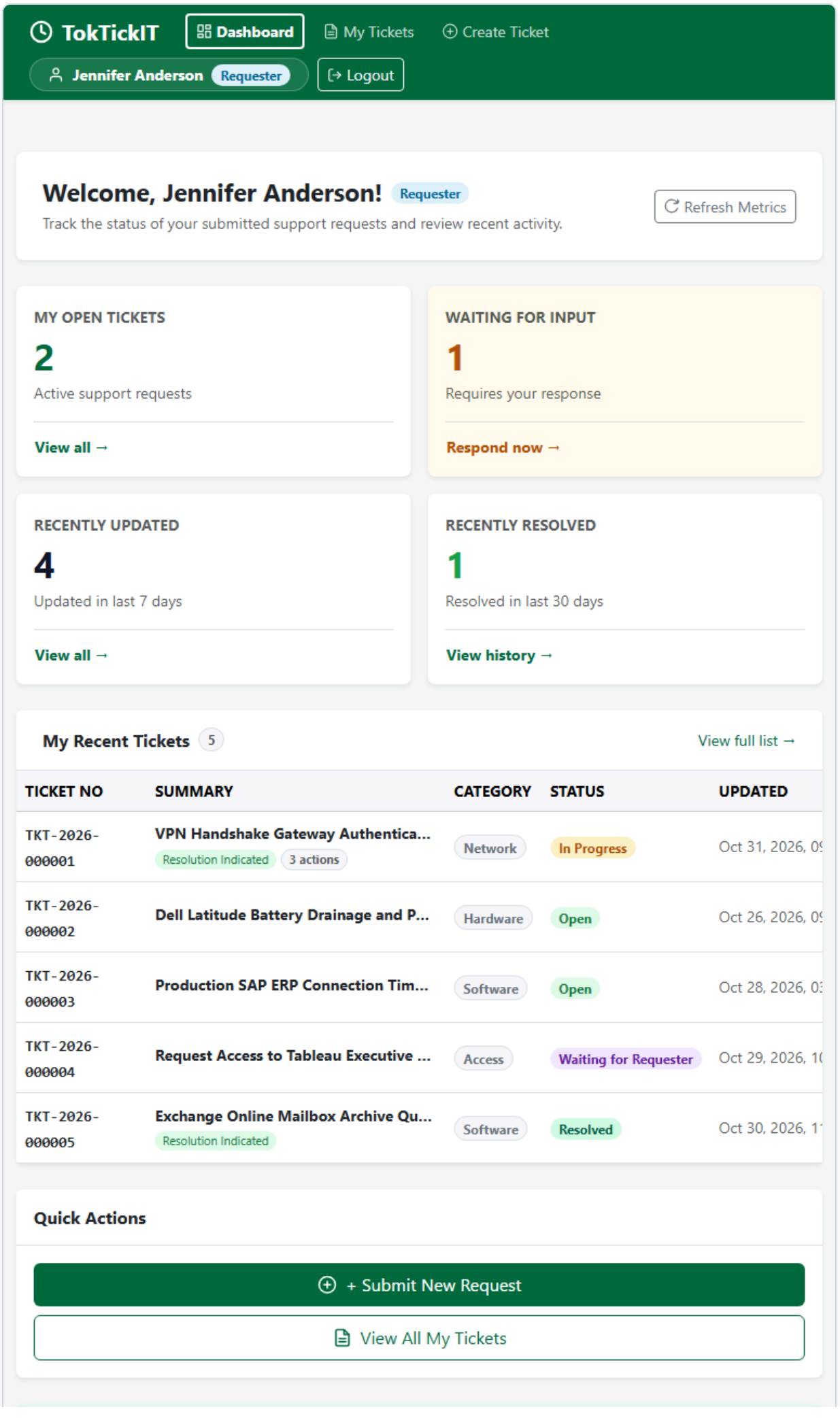
- **Welcome Banner**: Displays personalized greeting and user role.
- **4 Metric Cards**: My Open Tickets, Waiting for Input, Recently Updated, and Recently Resolved.
- **Recent Tickets Table**: Summarizes active requests with direct view links.
- **Quick Actions**: One-click ticket creation and queue access.

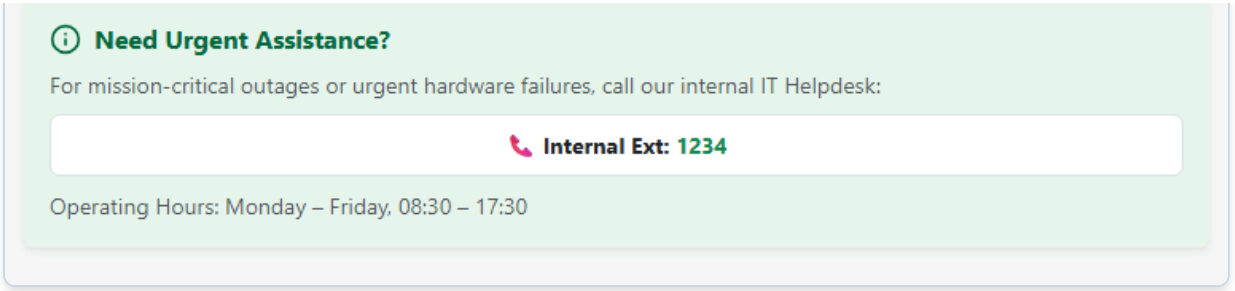
Requester Dashboard — Desktop (1280px)



Requester Dashboard Desktop

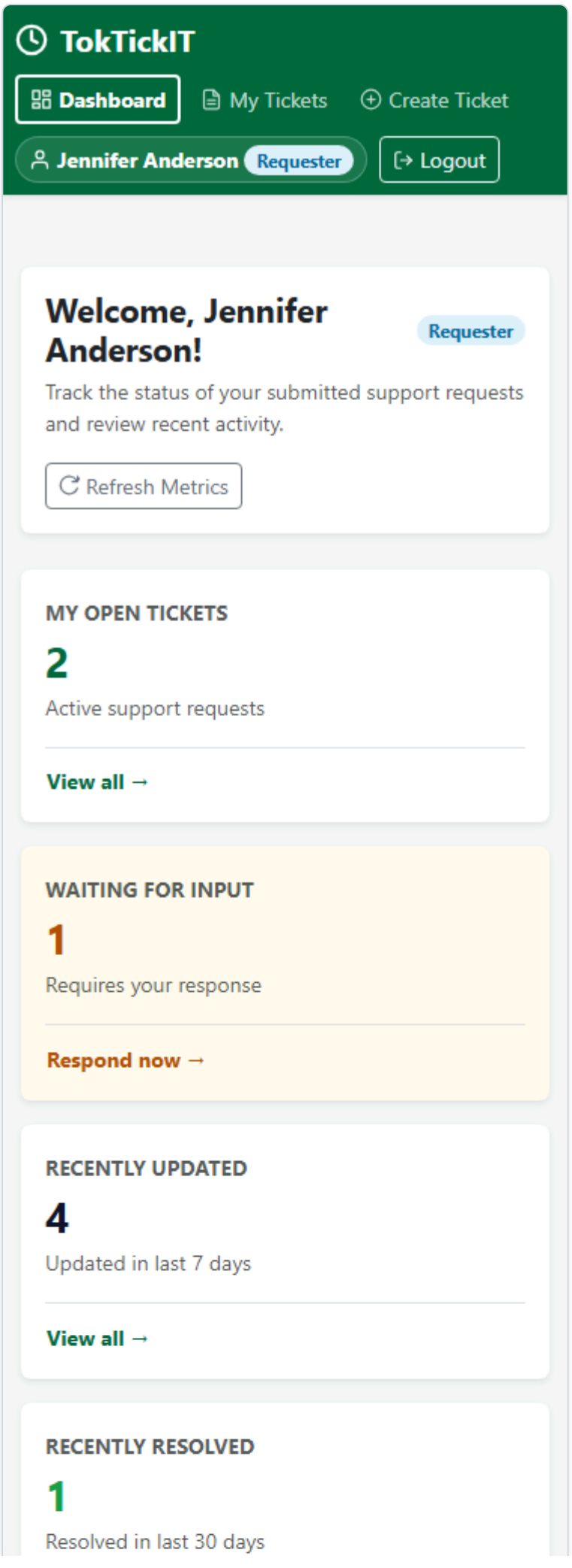
Requester Dashboard — Tablet (768px)

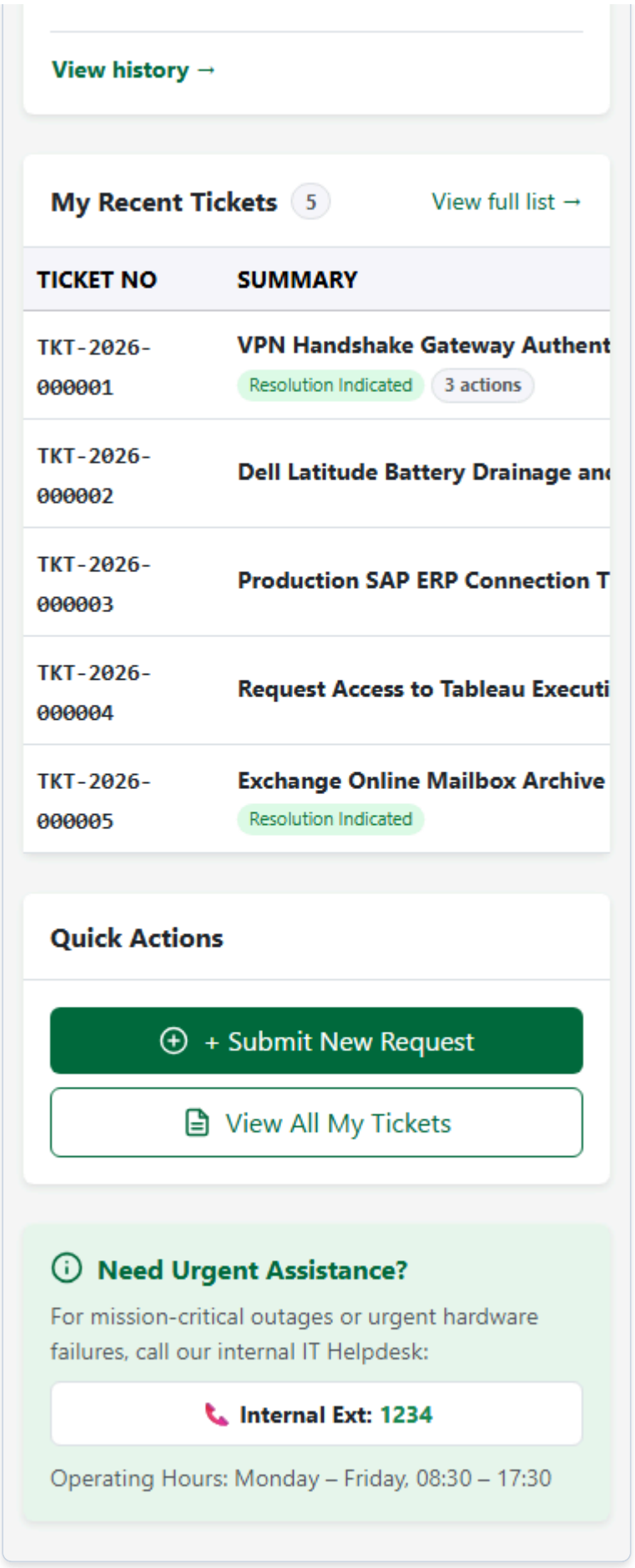

TokTickIT
Dashboard
My Tickets
Create Ticket
Jennifer Anderson
Requester
Logout
Welcome, Jennifer Anderson!
Requester
Track the status of your submitted support requests and review recent activity.
Refresh Metrics
MY OPEN TICKETS
2
Active support requests
View all →
WAITING FOR INPUT
1
Requires your response
Respond now →
RECENTLY UPDATED
4
Updated in last 7 days
View all →
RECENTLY RESOLVED
1
Resolved in last 30 days
View history →
My Recent Tickets
5
View full list →
TICKET NO
SUMMARY
CATEGORY
STATUS
UPDATED
TKT-2026-000001
VPN Handshake Gateway Authentica...
Resolution Indicated
3 actions
Network
In Progress
Oct 31, 2026, 0
TKT-2026-000002
Dell Latitude Battery Drainage and P...
Hardware
Open
Oct 26, 2026, 0
TKT-2026-000003
Production SAP ERP Connection Tim...
Software
Open
Oct 28, 2026, 0
TKT-2026-000004
Request Access to Tableau Executive ...
Access
Waiting for Requester
Oct 29, 2026, 1
TKT-2026-000005
Exchange Online Mailbox Archive Qu...
Resolution Indicated
Software
Resolved
Oct 30, 2026, 1
Quick Actions
+ Submit New Request
View All My Tickets



Requester Dashboard Tablet

Requester Dashboard — Mobile (375px)

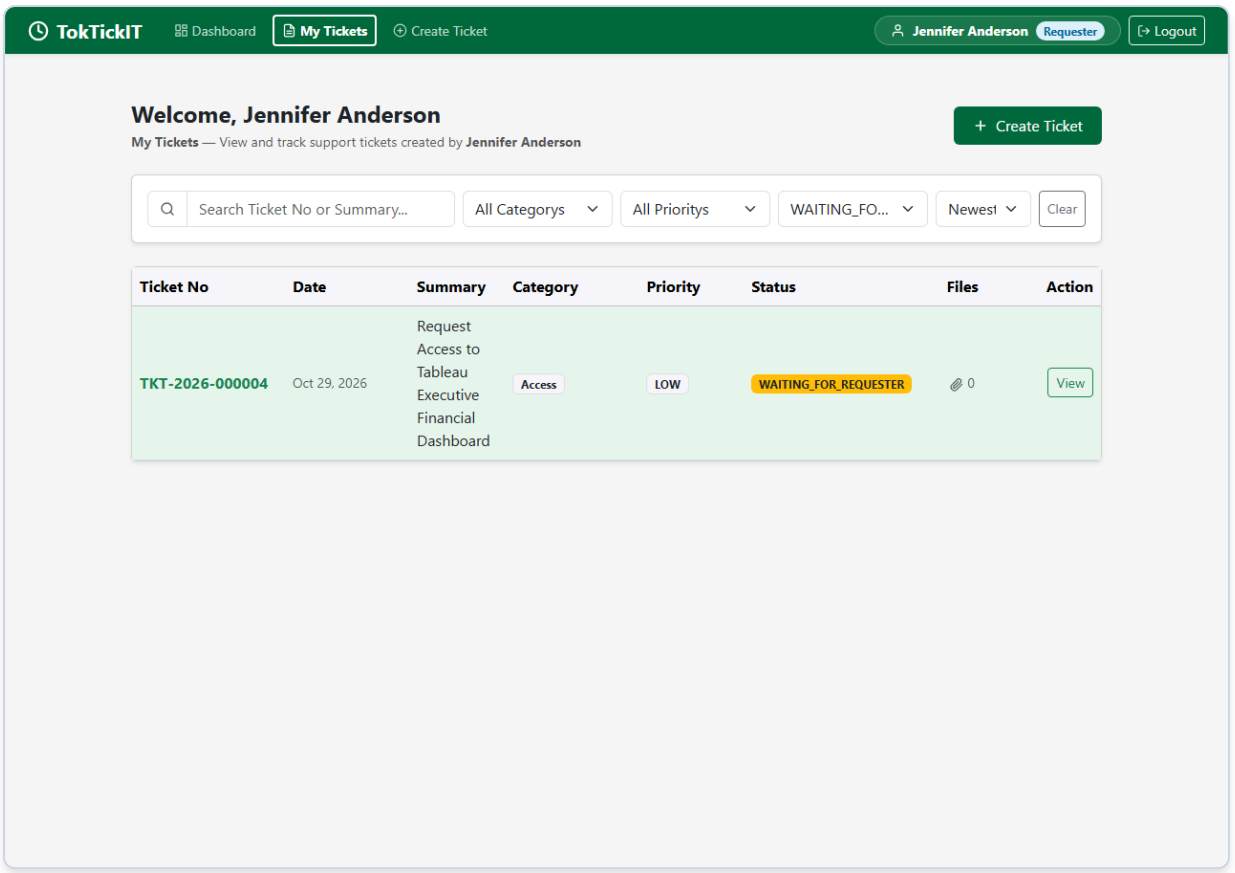

TokTickIT
Dashboard
My Tickets
Create Ticket
Jennifer Anderson
Requester
Logout
Welcome, Jennifer Anderson!
Requester
Track the status of your submitted support requests and review recent activity.
Refresh Metrics
MY OPEN TICKETS
2
Active support requests
View all →
WAITING FOR INPUT
1
Requires your response
Respond now →
RECENTLY UPDATED
4
Updated in last 7 days
View all →
RECENTLY RESOLVED
1
Resolved in last 30 days



Requester Dashboard Mobile

5.2 Requester Drill-Down Flow & Filtering Evidence

Clicking the "Waiting for Input" metric card automatically navigates to `MyTicketsView` pre-filtered by `WAITING_FOR_REQUESTER`, enabling immediate response without manual filter selection:



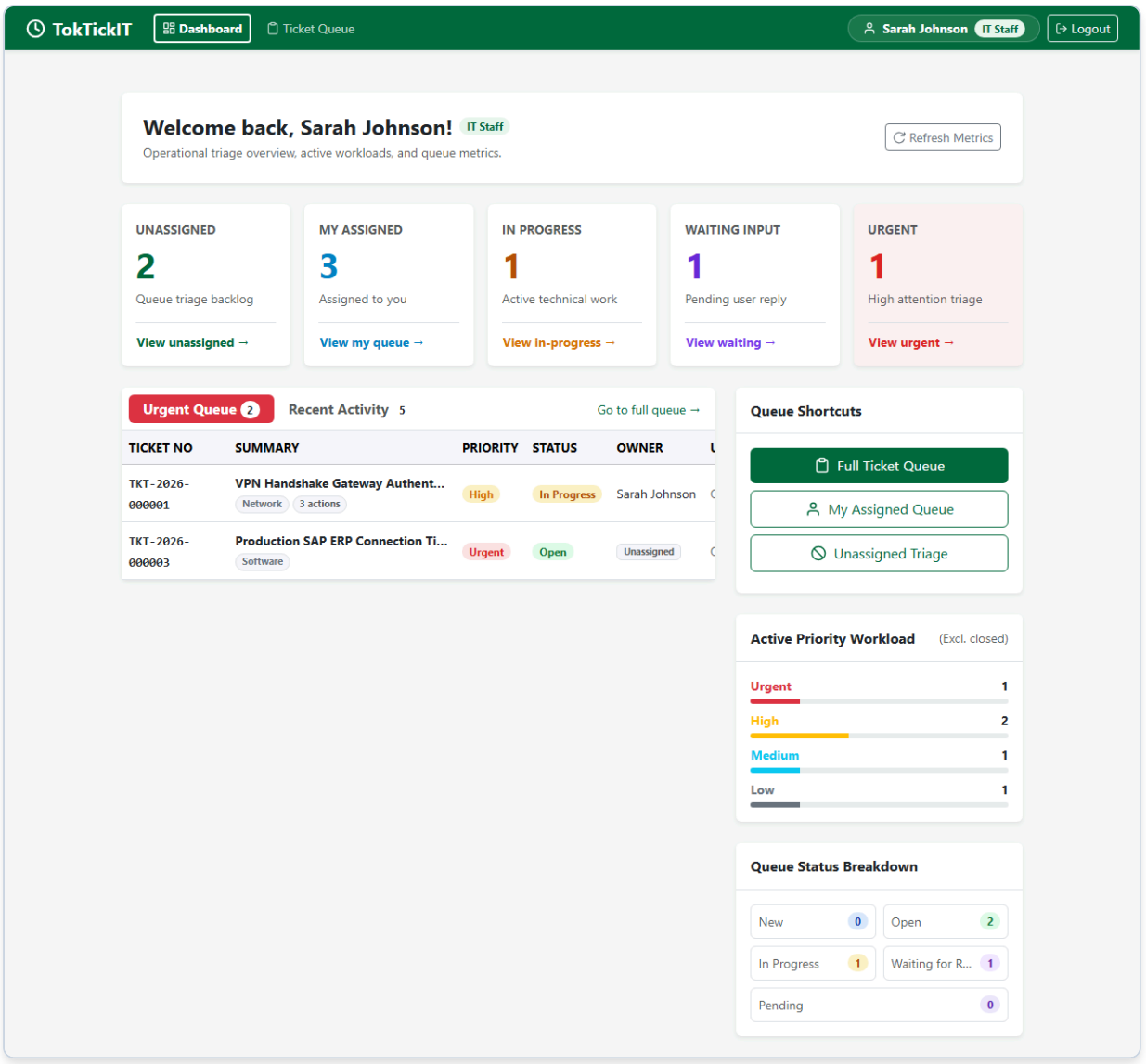
Requester Drill-Down Filtered

5.3 IT Staff Dashboard

The IT Staff Dashboard centralizes operational queue health and triage metrics:

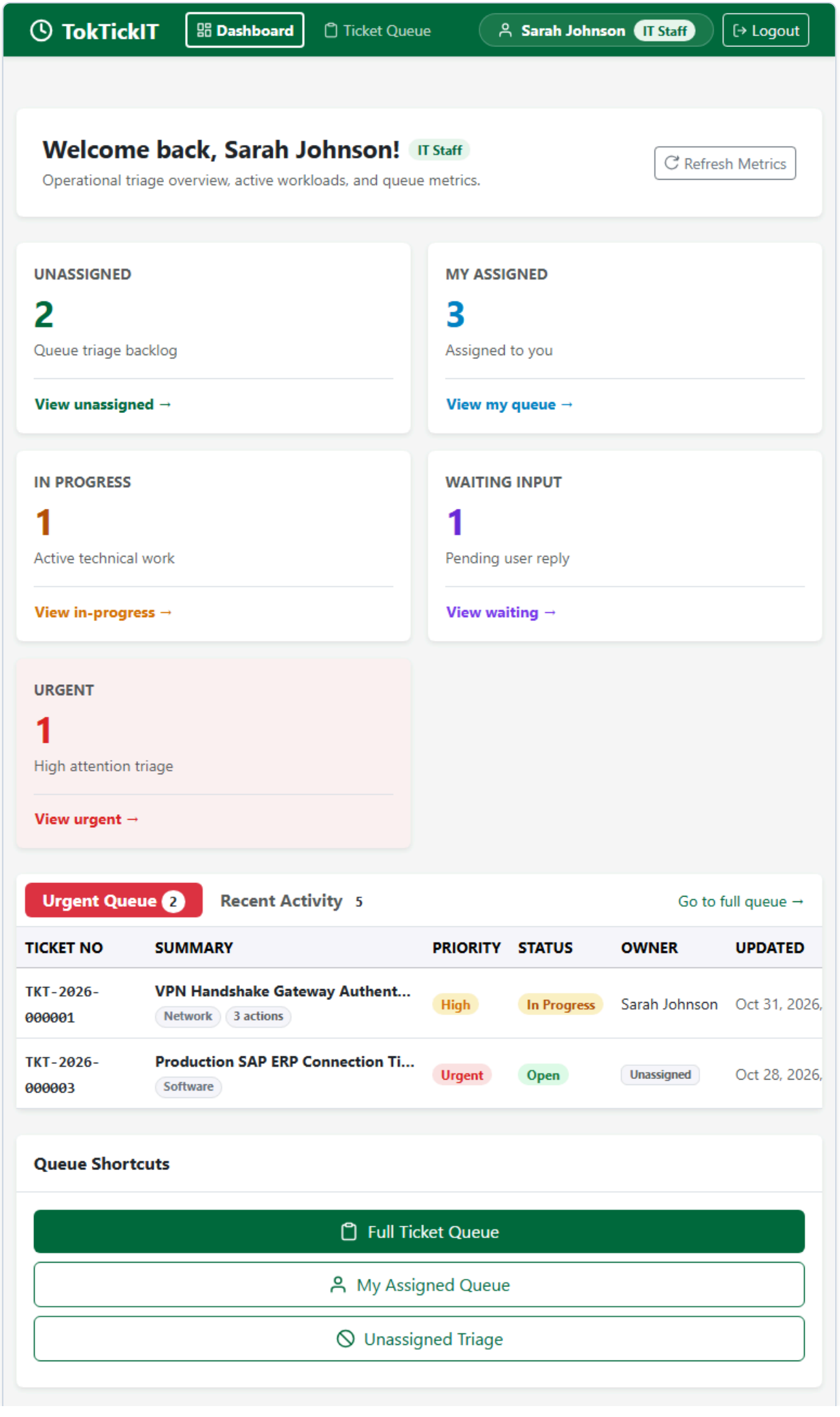
- **Operational Banner**: Summarizes active queue backlog.
- **Primary Metric Cards**: Unassigned Tickets, My Assigned Tickets, In Progress, Waiting for Requester, and Urgent Tickets.
- **Status & Priority Breakdowns**: Compact distribution chips for immediate workload balancing.
- **Urgent & Recent Tabs**: Priority triage table with direct ticket access.
- **Administrator System Stats**: Displays user statistics when viewed by Administrators.

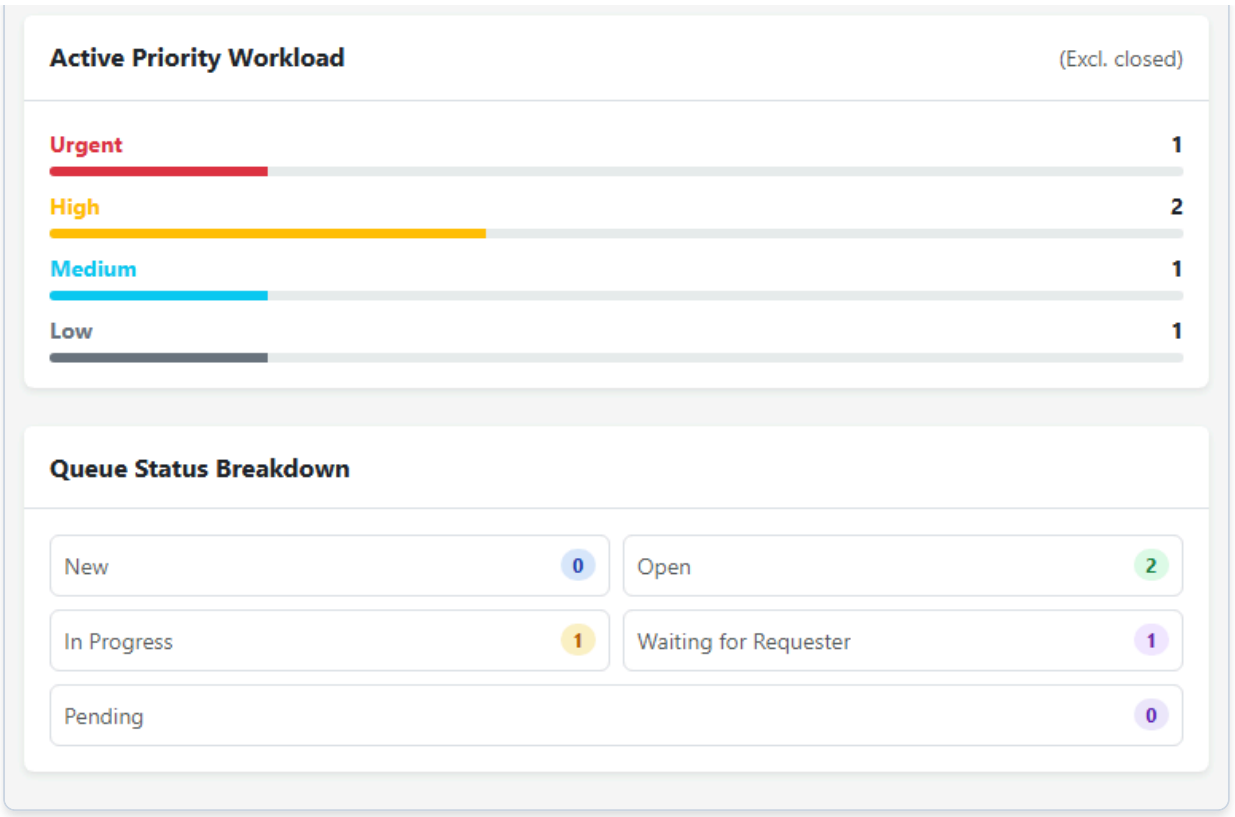
IT Staff Dashboard — Desktop (1280px)



IT Staff Dashboard Desktop

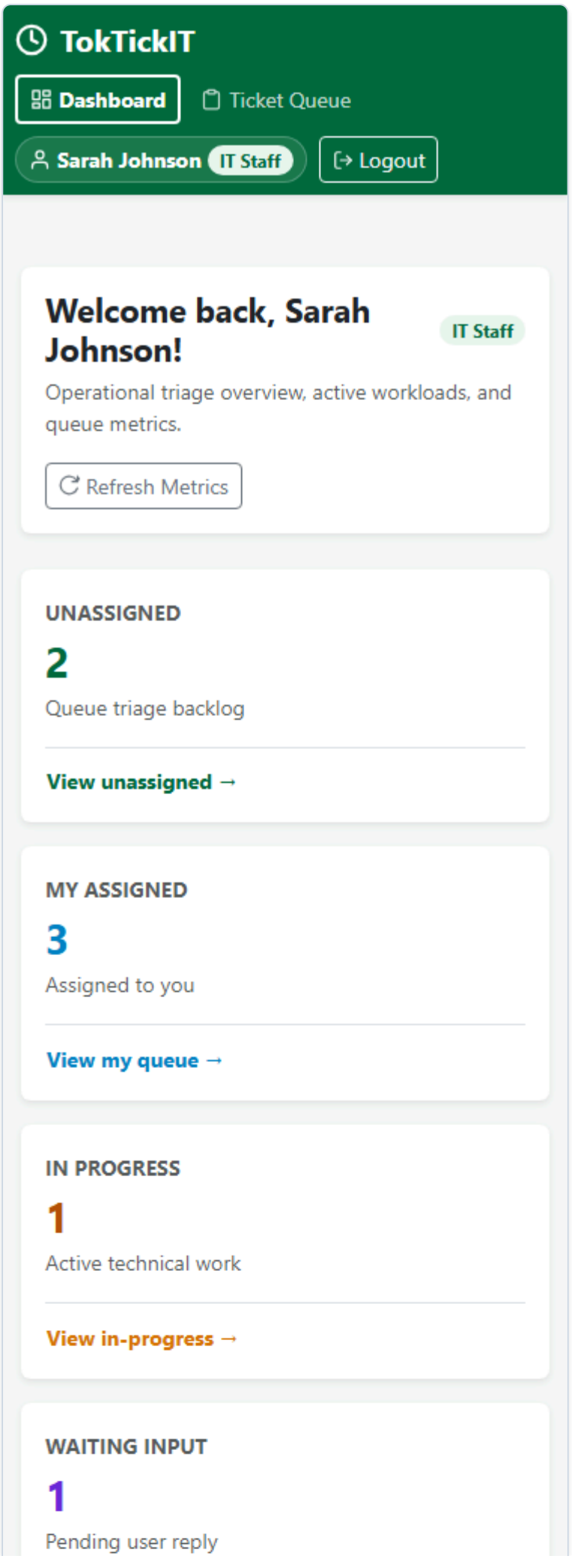
IT Staff Dashboard — Tablet (768px)


TokTickIT
Dashboard
Ticket Queue
Sarah Johnson
IT Staff
Logout
Welcome back, Sarah Johnson!
IT Staff
Operational triage overview, active workloads, and queue metrics.
Refresh Metrics
UNASSIGNED
2
Queue triage backlog
View unassigned →
MY ASSIGNED
3
Assigned to you
View my queue →
IN PROGRESS
1
Active technical work
View in-progress →
WAITING INPUT
1
Pending user reply
View waiting →
URGENT
1
High attention triage
View urgent →
Urgent Queue 2
Recent Activity 5
Go to full queue →
TICKET NO
SUMMARY
PRIORITY
STATUS
OWNER
UPDATED
TKT-2026-000001
VPN Handshake Gateway Authent...
Network
3 actions
High
In Progress
Sarah Johnson
Oct 31, 2026,
TKT-2026-000003
Production SAP ERP Connection Ti...
Software
Urgent
Open
Unassigned
Oct 28, 2026,
Queue Shortcuts
Full Ticket Queue
My Assigned Queue
Unassigned Triage



IT Staff Dashboard Tablet

IT Staff Dashboard — Mobile (375px)


TokTickIT
Dashboard
Ticket Queue
Sarah Johnson
IT Staff
Logout
Welcome back, Sarah Johnson!
IT Staff
Operational triage overview, active workloads, and queue metrics.
Refresh Metrics
UNASSIGNED
2
Queue triage backlog
View unassigned →
MY ASSIGNED
3
Assigned to you
View my queue →
IN PROGRESS
1
Active technical work
View in-progress →
WAITING INPUT
1
Pending user reply

View waiting →

URGENT

1

High attention triage

View urgent →

Urgent Queue 2

Recent Activity 5

Go to full queue →

TICKET NO	SUMMARY
TKT-2026-000001	VPN Handshake Gateway Authent Network 3 actions
TKT-2026-000003	Production SAP ERP Connection T Software

Queue Shortcuts

Full Ticket Queue

My Assigned Queue

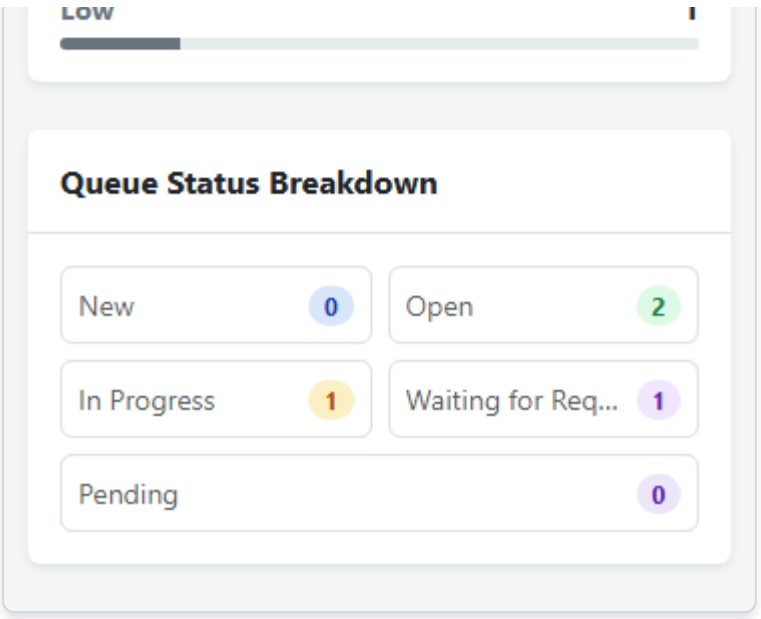
Unassigned Triage

Active Priority Workload (Excl. closed)

Urgent 1

High 2

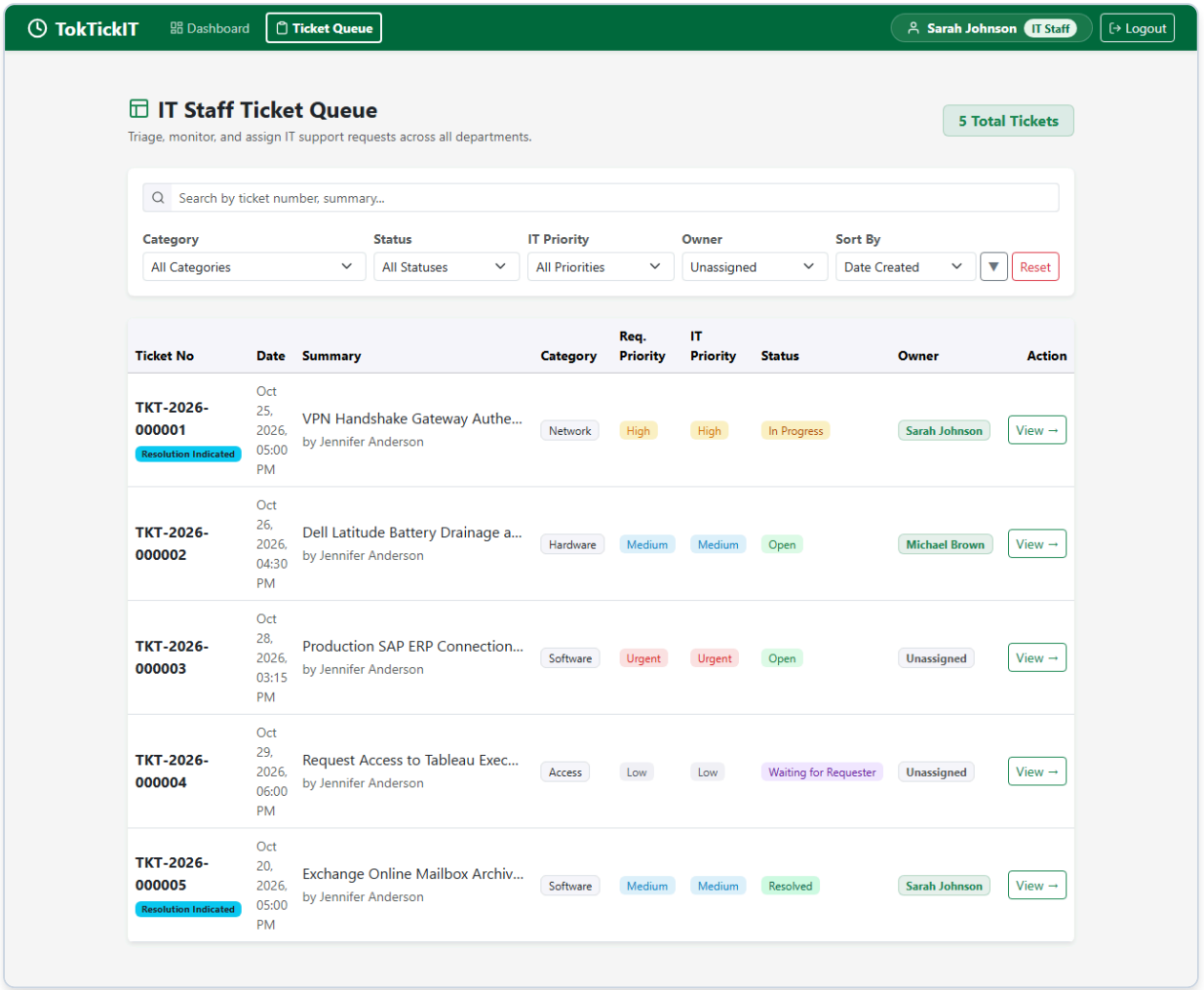
Medium 1



IT Staff Dashboard Mobile

5.4 IT Staff Queue Drill-Down & Triage Queue Evidence

Clicking the "Unassigned Tickets" card drills down into `StaffTicketQueue` pre-filtered for unassigned items, allowing instant batch claiming:



IT Staff Drill-Down Queue

Answer Part 6: Working Actions Taken UI

6.1 Actions Taken Technical Architecture & Lifecycle

Actions Taken represent concrete engineering steps under each ticket. Each action follows a strict lifecycle: `PENDING` -> `IN_PROGRESS` -> `COMPLETED` or `CANCELLED`. Performer attribution is automatically assigned to the authenticated user, while assignee assignment is restricted to active staff accounts.

6.2 Desktop Table View & Mobile Responsive Cards

Actions Taken Desktop Table (1280px)

Displays Date/Time, Description, Result/Outcome, Performed By, Assigned To, Status Badge, Follow-Up badge, and Action buttons:

TokTickIT | Dashboard | Ticket Queue | Sarah Johnson IT Staff | Logout

Ticket Queue / **Ticket Details**

TKT-2026-000001 Resolution Indicated

← Back to Ticket Queue

Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.

Ticket No.	Ticket Date	Category	Related System
TKT-2026-000001	Oct 25, 2026, 05:00 PM	Network	GlobalProtect VPN

Requester	Requested Priority	IT Priority	Current Status
Jennifer Anderson	Priority: HIGH	Priority: HIGH	Status: IN_PROGRESS

Ticket Owner	Summary
Sarah Johnson	VPN Handshake Gateway Authentication Failure

Description

After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.

Resolution Summary

No resolution summary available yet.

IT Staff Ticket Operations Staff Control

Quick Claim	Assign Owner	IT Priority	Change Status
Claimed by You	Sarah Johnson (Staff)	HIGH	Select Next Status... Apply

Actions Taken by IT Staff 3 Actions Recorded + Add Action Taken

Date / Time	Description	Result / Outcome	Performed By	Assigned To	Status	Follow-Up	Attachm
25 Oct 2026, 18:00	Inspected network gateway RADIUS logs and validated intermediate CA certificate revocation list	Discovered expired intermediate certificate CRL cache on client machine	Sarah Johnson IT Staff	Michael Brown IT Staff	Completed	*None*	RADI
25 Oct 2026, 21:30	Pushed group policy certificate store refresh and tested client tunnel establishment	Tunnel re-established successfully; monitored connection latency for 30 minutes	Michael Brown IT Staff	Sarah Johnson IT Staff	Completed	Yes	gpo-r
31 Oct 2026, 16:00	Final gateway validation and stability verification under sustained VPN traffic	—	Sarah Johnson IT Staff	Sarah Johnson IT Staff	In Progress	*None*	—

Public Comments 2 | Internal Notes 1 | Attachments | Service Actions 3 | Event Log 6

Public Comments — Visible to Requester 2

Sarah Johnson IT Staff — Oct 25, 2026, 05:30 PM

We have reviewed the network gateway logs. The certificate revocation list (CRL) cache on client side requires a force refresh.

Jennifer Anderson Requester — Oct 31, 2026, 09:30 PM

I refreshed the certificate store and connected successfully. ERP access is functioning normally now!

Add Public Comment

Type a public comment (visible to requester and staff)...

0 / 2000 characters

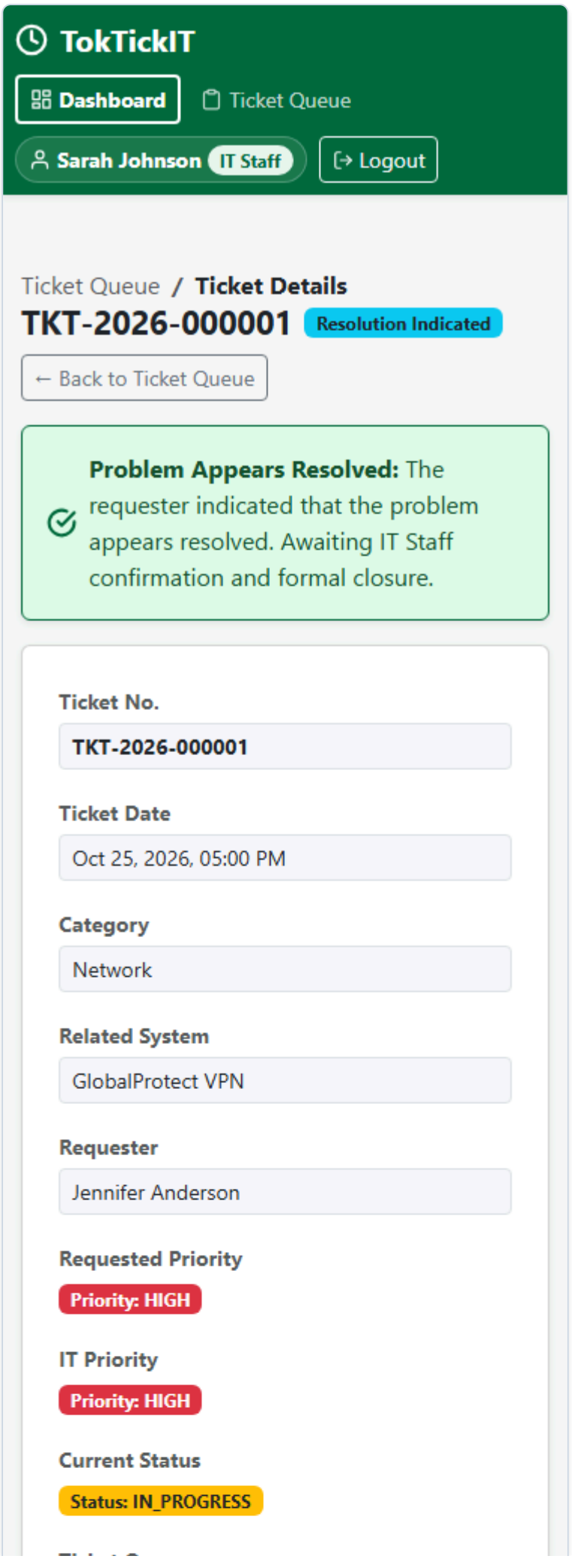
Post Comment

« Back to Ticket Queue

Actions Taken Desktop Table

Actions Taken Mobile Responsive Cards (375px)

On mobile devices (< 768px), the table seamlessly transforms into touch-optimized cards without horizontal clipping:


TokTickIT
Dashboard
Ticket Queue
Sarah Johnson
IT Staff
Logout
Ticket Queue / Ticket Details
TKT-2026-000001
Resolution Indicated
← Back to Ticket Queue
Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.
Ticket No.
TKT-2026-000001
Ticket Date
Oct 25, 2026, 05:00 PM
Category
Network
Related System
GlobalProtect VPN
Requester
Jennifer Anderson
Requested Priority
Priority: HIGH
IT Priority
Priority: HIGH
Current Status
Status: IN_PROGRESS

Ticket Owner

Sarah Johnson

Summary

VPN Handshake Gateway Authentication Failure

Description

After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.

Resolution Summary

No resolution summary available yet.

IT Staff Ticket Operations Staff Control

Quick Claim

Claimed by You

Assign Owner

Sarah Johnson (Staff)

IT Priority

HIGH

Change Status

Select Next Status... Apply

Actions Taken by IT Staff 3 Actions Recorded

+ Add Action Taken

25 Oct 2026, 18:00 Completed

Inspected network gateway RADIUS logs and validated intermediate CA

certificate revocation list

Outcome: Discovered expired intermediate certificate CRL cache on client machine

By: Sarah Johnson •

Assignee: Michael Brown

RADIUS-log-dump-20261025.txt

25 Oct 2026, 21:30 Completed

Pushed group policy certificate store refresh and tested client tunnel establishment

Outcome: Tunnel re-established successfully; monitored connection latency for 30 minutes

By: Michael Brown •

Assignee: Sarah Johnson

Follow-up: Verify certificate auto-enrollment persists through next scheduled domain sync

gpo-result-report.pdf

31 Oct 2026, 16:00 In Progress

Final gateway validation and stability verification under sustained VPN traffic

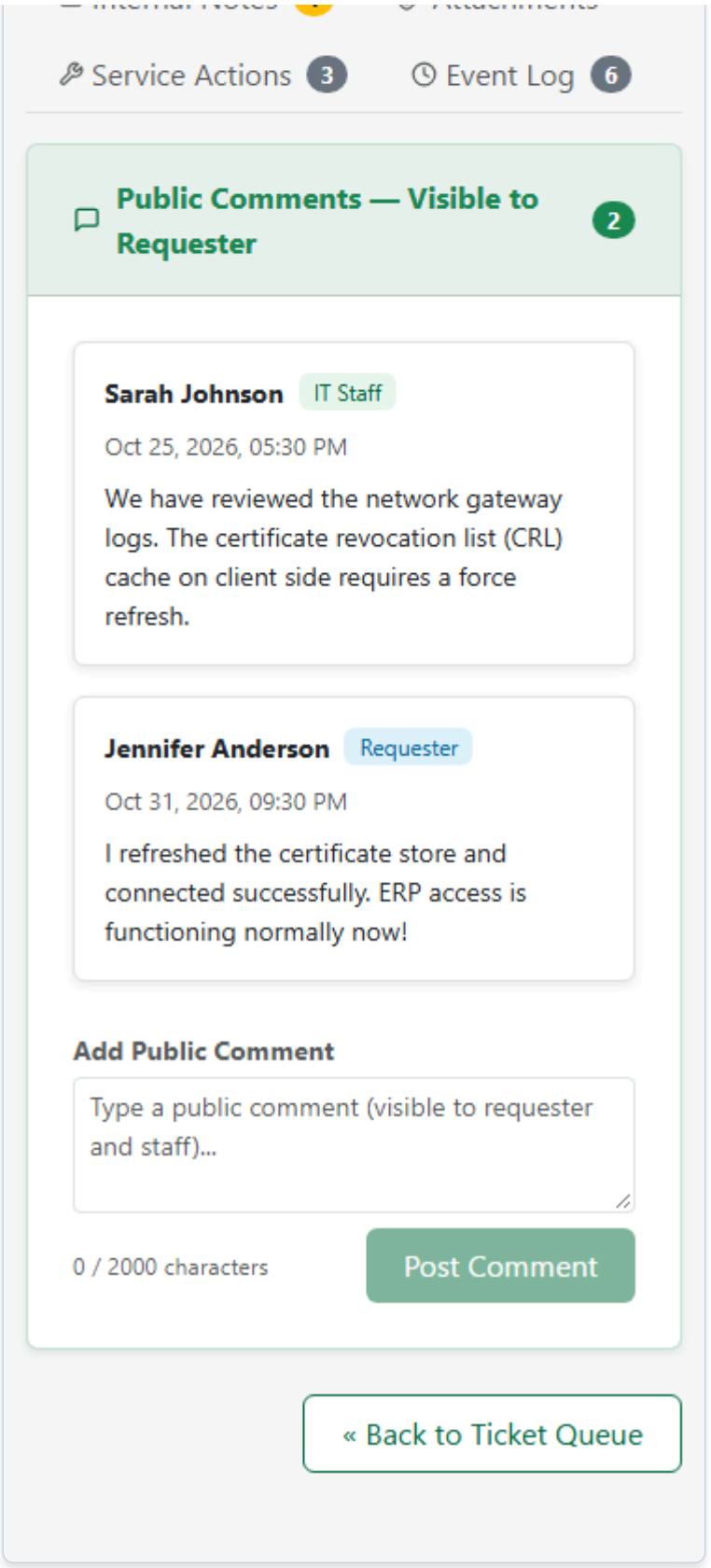
By: Sarah Johnson •

Assignee: Sarah Johnson

Edit Action Status ▾

Public Comments 2

Internal Notes 1 Attachments

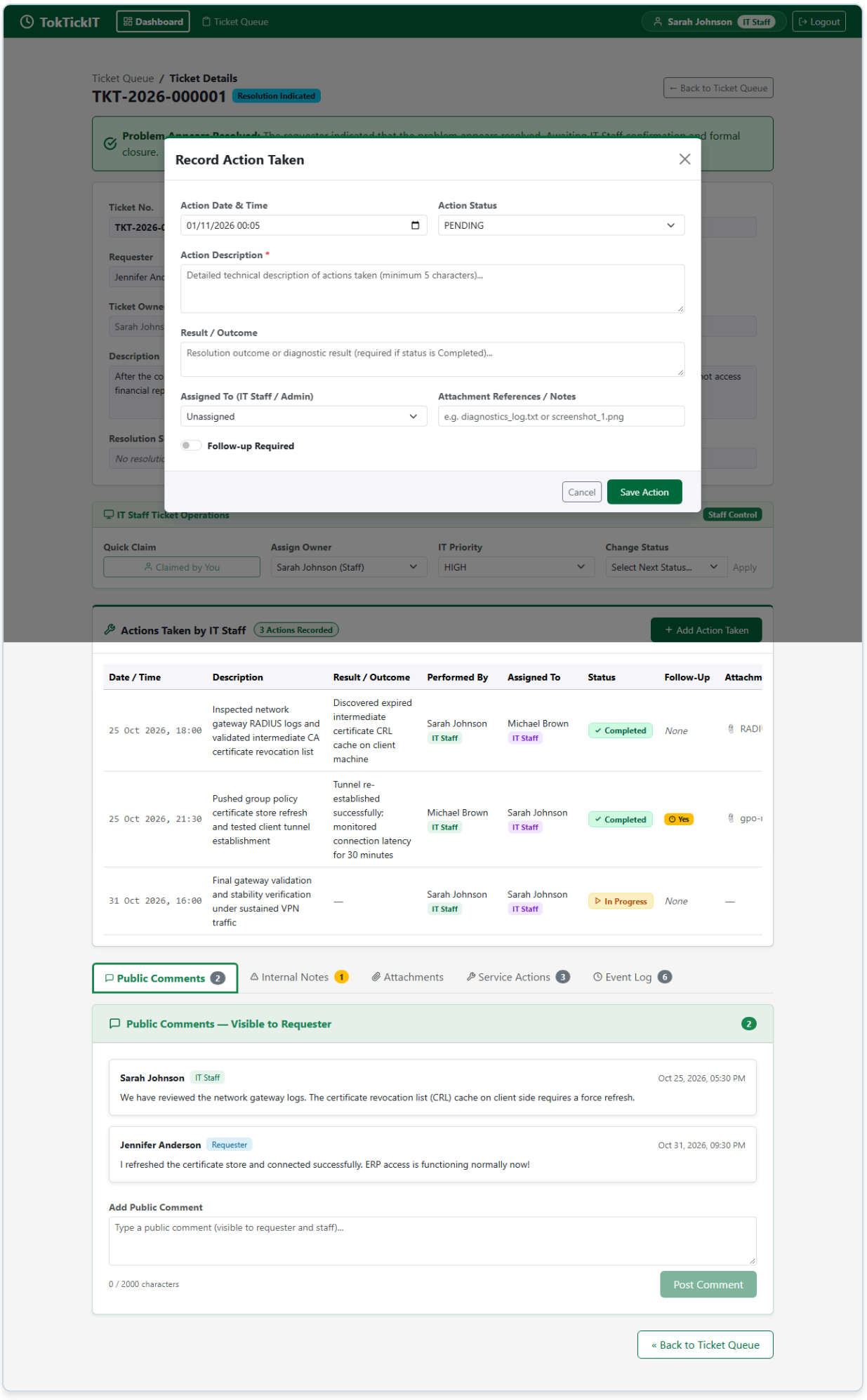


Actions Taken Mobile Cards

6.3 Add Action & Edit Action Modals

Create Action Modal

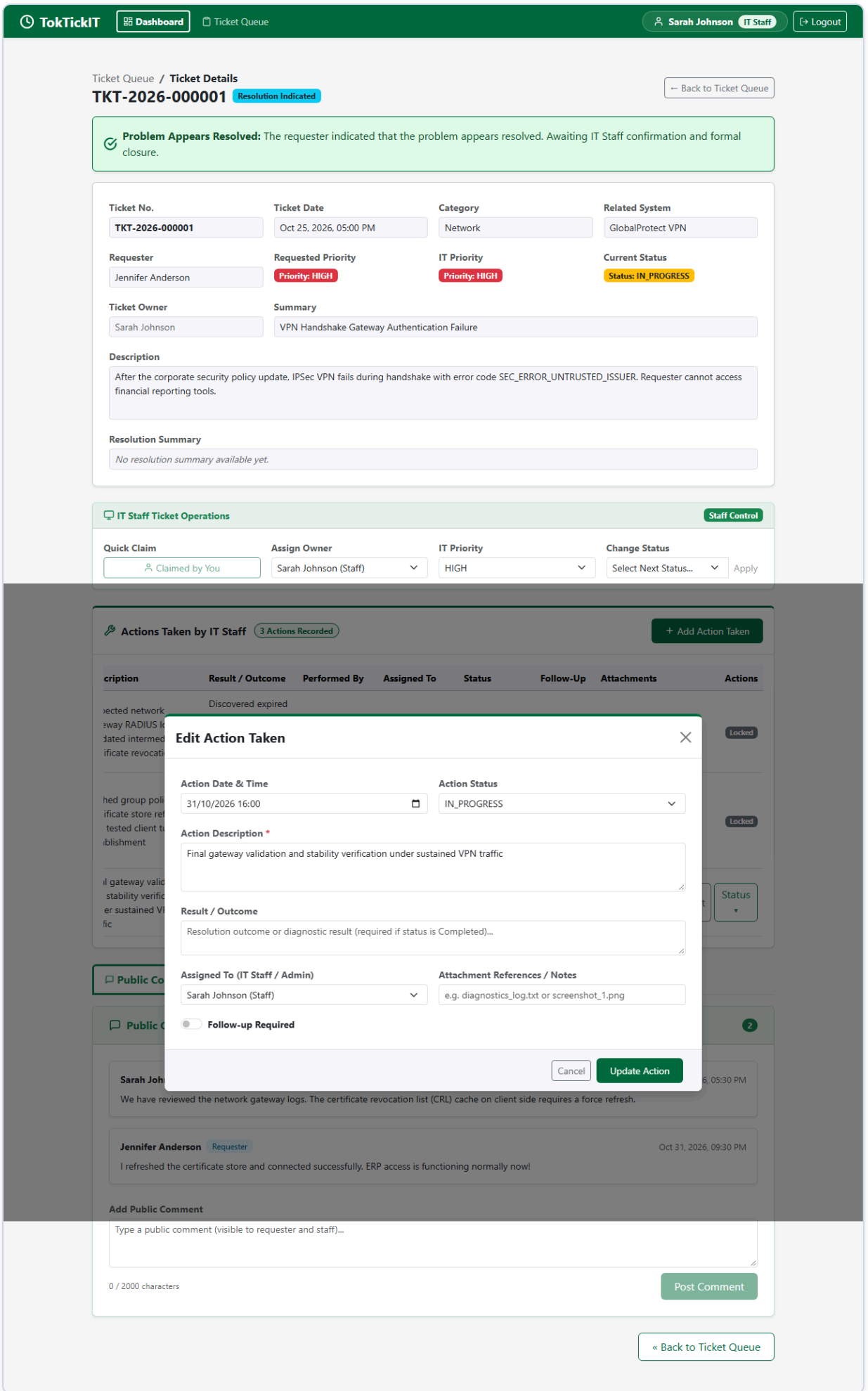
Supports date selection, description, outcome, assignee dropdown, status selection, and follow-up toggle:



Create Action Modal

Edit Action Modal

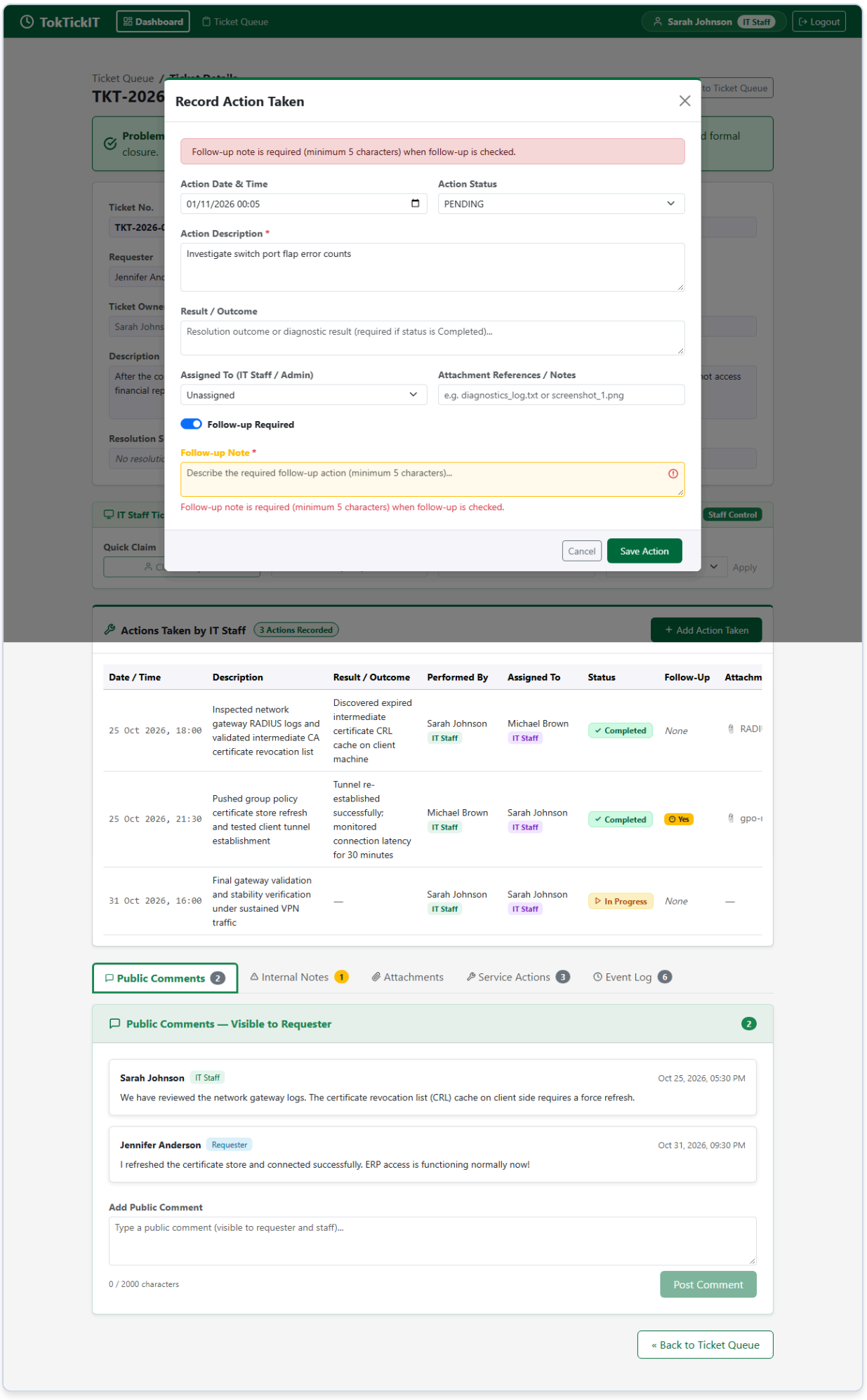
Allows updating in-progress actions with optimistic concurrency timestamp validation:



Edit Action Modal

6.4 Safe Failure 1: Follow-Up Required Validation Error

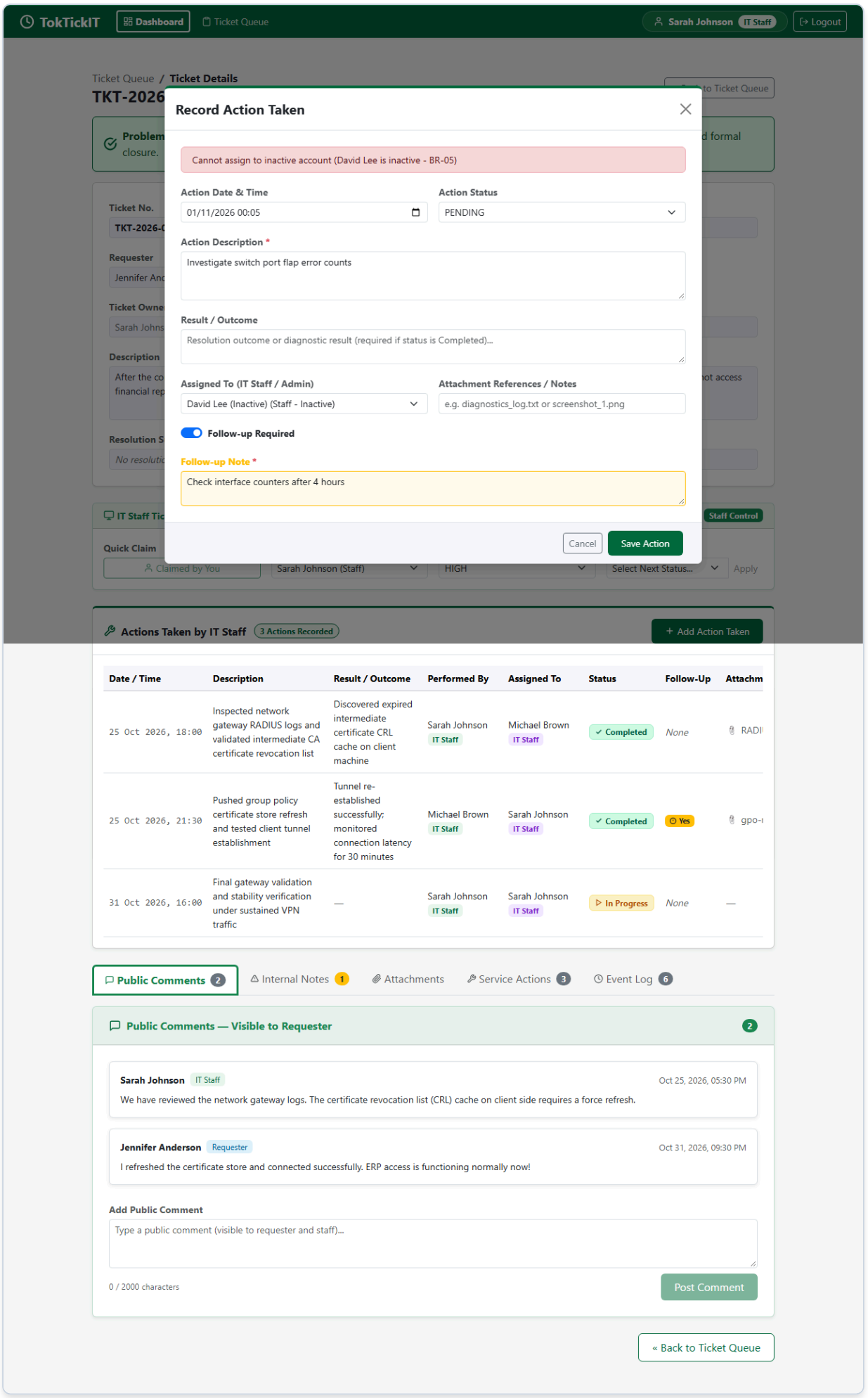
When the follow-up switch is enabled without providing a note (minimum 5 characters), form submission is blocked with inline validation:



Follow-up Required Error

6.5 Safe Failure 2: Inactive Assignee Guard Rejection (BR-05)

Selecting an inactive staff account (e.g. David Lee - Inactive) triggers server validation rejection with HTTP 422 Unprocessable Entity:



Inactive Assignee Rejection

Answer Part 7: Working Resolution Gate & Ticket Workflow UI

7.1 Advisory Resolution Indication Banner & Flow

When a requester reports that their problem appears resolved, a prominent Zen Green banner alerts IT Staff while keeping the ticket active for formal review:

TokTickIT | Dashboard | Ticket Queue | Sarah Johnson IT Staff | Logout

Ticket Queue / **Ticket Details**

TKT-2026-000001 Resolution Indicated

← Back to Ticket Queue

Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.

Ticket No.	Ticket Date	Category	Related System
TKT-2026-000001	Oct 25, 2026, 05:00 PM	Network	GlobalProtect VPN

Requester	Requested Priority	IT Priority	Current Status
Jennifer Anderson	Priority: HIGH	Priority: HIGH	Status: IN_PROGRESS

Ticket Owner	Summary
Sarah Johnson	VPN Handshake Gateway Authentication Failure

Description

After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.

Resolution Summary

No resolution summary available yet.

IT Staff Ticket Operations Staff Control

Quick Claim	Assign Owner	IT Priority	Change Status
Claimed by You	Sarah Johnson (Staff)	HIGH	Select Next Status... Apply

Actions Taken by IT Staff 3 Actions Recorded + Add Action Taken

Date / Time	Description	Result / Outcome	Performed By	Assigned To	Status	Follow-Up	Attachm
25 Oct 2026, 18:00	Inspected network gateway RADIUS logs and validated intermediate CA certificate revocation list	Discovered expired intermediate certificate CRL cache on client machine	Sarah Johnson IT Staff	Michael Brown IT Staff	Completed	*None*	RADI
25 Oct 2026, 21:30	Pushed group policy certificate store refresh and tested client tunnel establishment	Tunnel re-established successfully; monitored connection latency for 30 minutes	Michael Brown IT Staff	Sarah Johnson IT Staff	Completed	Yes	gpo-r
31 Oct 2026, 16:00	Final gateway validation and stability verification under sustained VPN traffic	—	Sarah Johnson IT Staff	Sarah Johnson IT Staff	In Progress	*None*	—

Public Comments 2 | Internal Notes 1 | Attachments | Service Actions 3 | Event Log 6

Public Comments — Visible to Requester 2

Sarah Johnson IT Staff — Oct 25, 2026, 05:30 PM

We have reviewed the network gateway logs. The certificate revocation list (CRL) cache on client side requires a force refresh.

Jennifer Anderson Requester — Oct 31, 2026, 09:30 PM

I refreshed the certificate store and connected successfully. ERP access is functioning normally now!

Add Public Comment

Type a public comment (visible to requester and staff)...

0 / 2000 characters

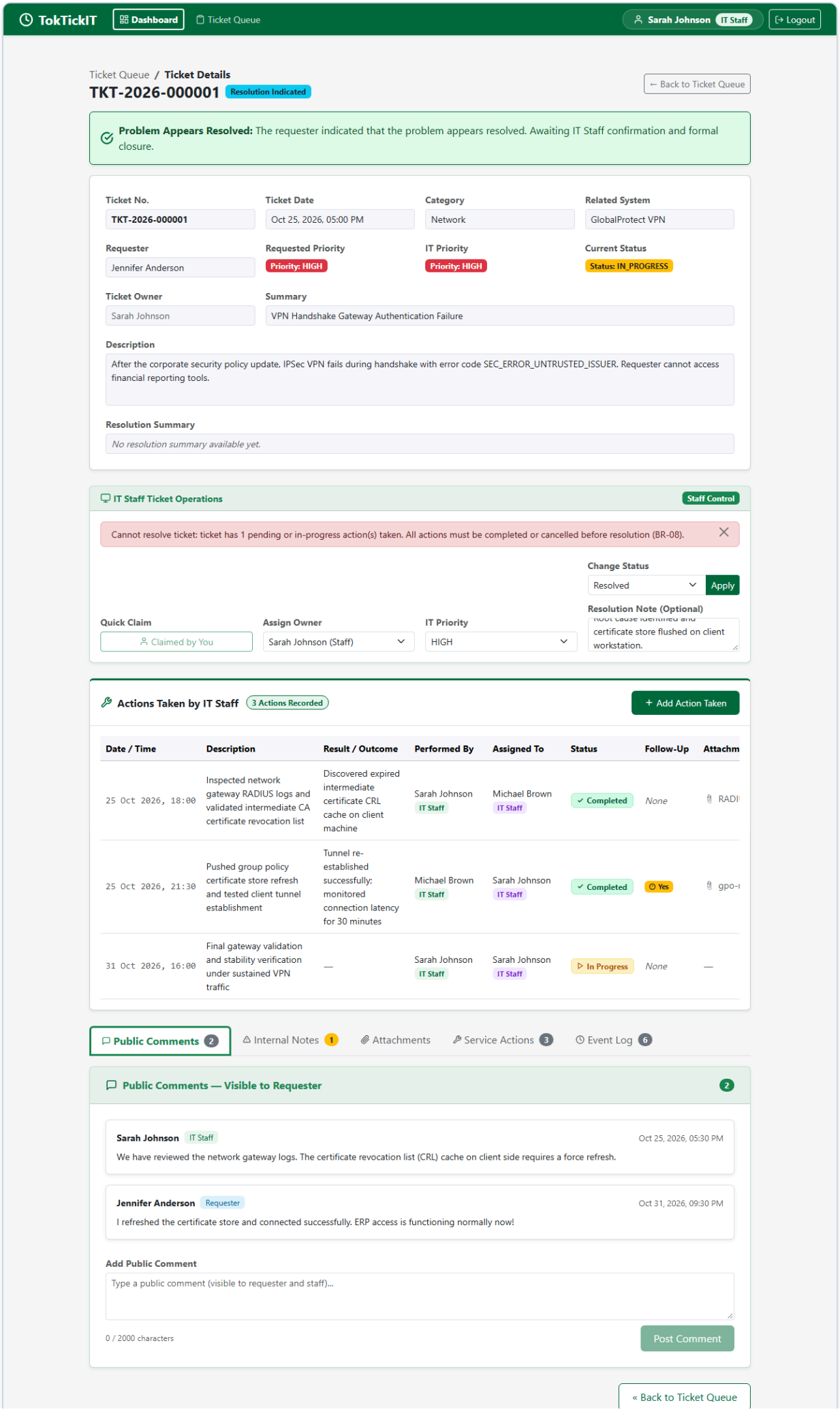
Post Comment

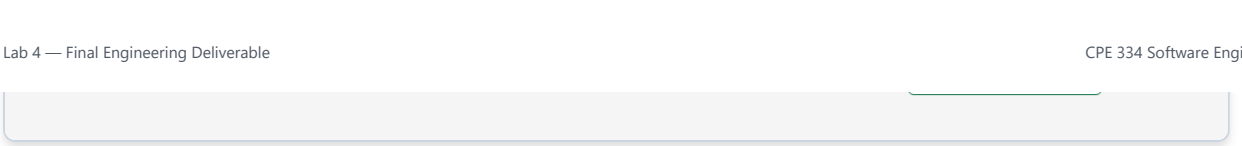
« Back to Ticket Queue

Requester Resolution Indicated Banner

7.2 Resolution Gate Precondition Enforcement (BR-08)

If IT Staff attempts to transition a ticket to `RESOLVED` while child actions remain in `PENDING` or `IN_PROGRESS`, the transition is blocked by the backend (HTTP 409 Conflict) and an error banner is displayed:

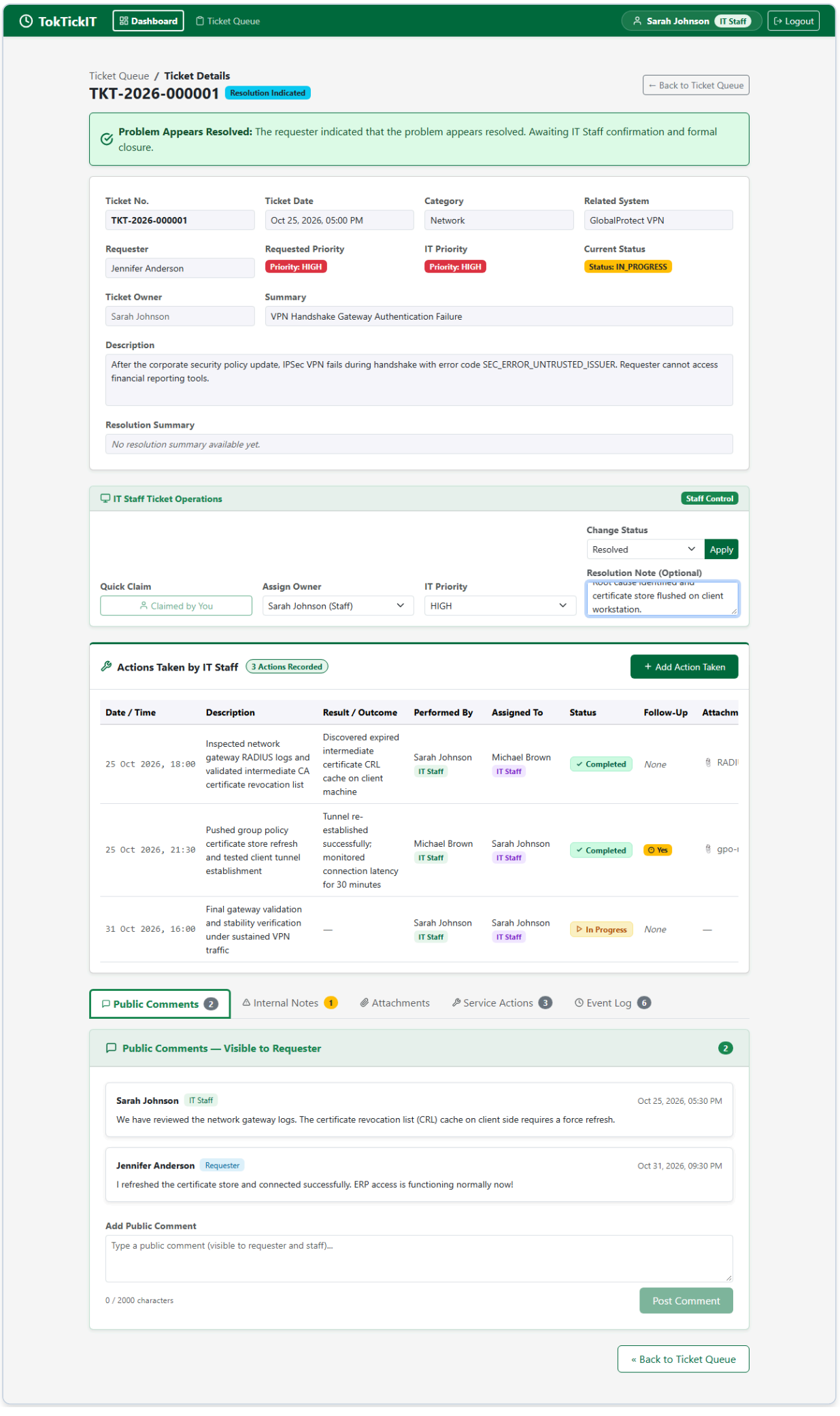

TokTickIT
Dashboard
Ticket Queue
Sarah Johnson IT Staff
Logout
Ticket Queue / Ticket Details
TKT-2026-000001 Resolution Indicated
← Back to Ticket Queue
Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.
Ticket No.
TKT-2026-000001
Ticket Date
Oct 25, 2026, 05:00 PM
Category
Network
Related System
GlobalProtect VPN
Requester
Jennifer Anderson
Requested Priority
Priority: HIGH
IT Priority
Priority: HIGH
Current Status
Status: IN_PROGRESS
Ticket Owner
Sarah Johnson
Summary
VPN Handshake Gateway Authentication Failure
Description
After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.
Resolution Summary
No resolution summary available yet.
IT Staff Ticket Operations
Staff Control
Cannot resolve ticket: ticket has 1 pending or in-progress action(s) taken. All actions must be completed or cancelled before resolution (BR-08).
Change Status
Resolved
Apply
Quick Claim
Claimed by You
Assign Owner
Sarah Johnson (Staff)
IT Priority
HIGH
Resolution Note (Optional)
certificate store flushed on client workstation.
Actions Taken by IT Staff
3 Actions Recorded
+ Add Action Taken
Date / Time
Description
Result / Outcome
Performed By
Assigned To
Status
Follow-Up
Attachm
25 Oct 2026, 18:00
Inspected network gateway RADIUS logs and validated intermediate CA certificate revocation list
Discovered expired intermediate certificate CRL cache on client machine
Sarah Johnson IT Staff
Michael Brown IT Staff
Completed
None
RADI
25 Oct 2026, 21:30
Pushed group policy certificate store refresh and tested client tunnel establishment
Tunnel re-established successfully; monitored connection latency for 30 minutes
Michael Brown IT Staff
Sarah Johnson IT Staff
Completed
Yes
gpo-
31 Oct 2026, 16:00
Final gateway validation and stability verification under sustained VPN traffic
—
Sarah Johnson IT Staff
Sarah Johnson IT Staff
In Progress
None
—
Public Comments 2
Internal Notes 1
Attachments
Service Actions 3
Event Log 6
Public Comments — Visible to Requester
2
Sarah Johnson IT Staff
Oct 25, 2026, 05:30 PM
We have reviewed the network gateway logs. The certificate revocation list (CRL) cache on client side requires a force refresh.
Jennifer Anderson Requester
Oct 31, 2026, 09:30 PM
I refreshed the certificate store and connected successfully. ERP access is functioning normally now!
Add Public Comment
Type a public comment (visible to requester and staff)...
0 / 2000 characters
Post Comment
« Back to Ticket Queue



Premature Resolution Blocked

7.3 IT Staff Resolution Note Input Field

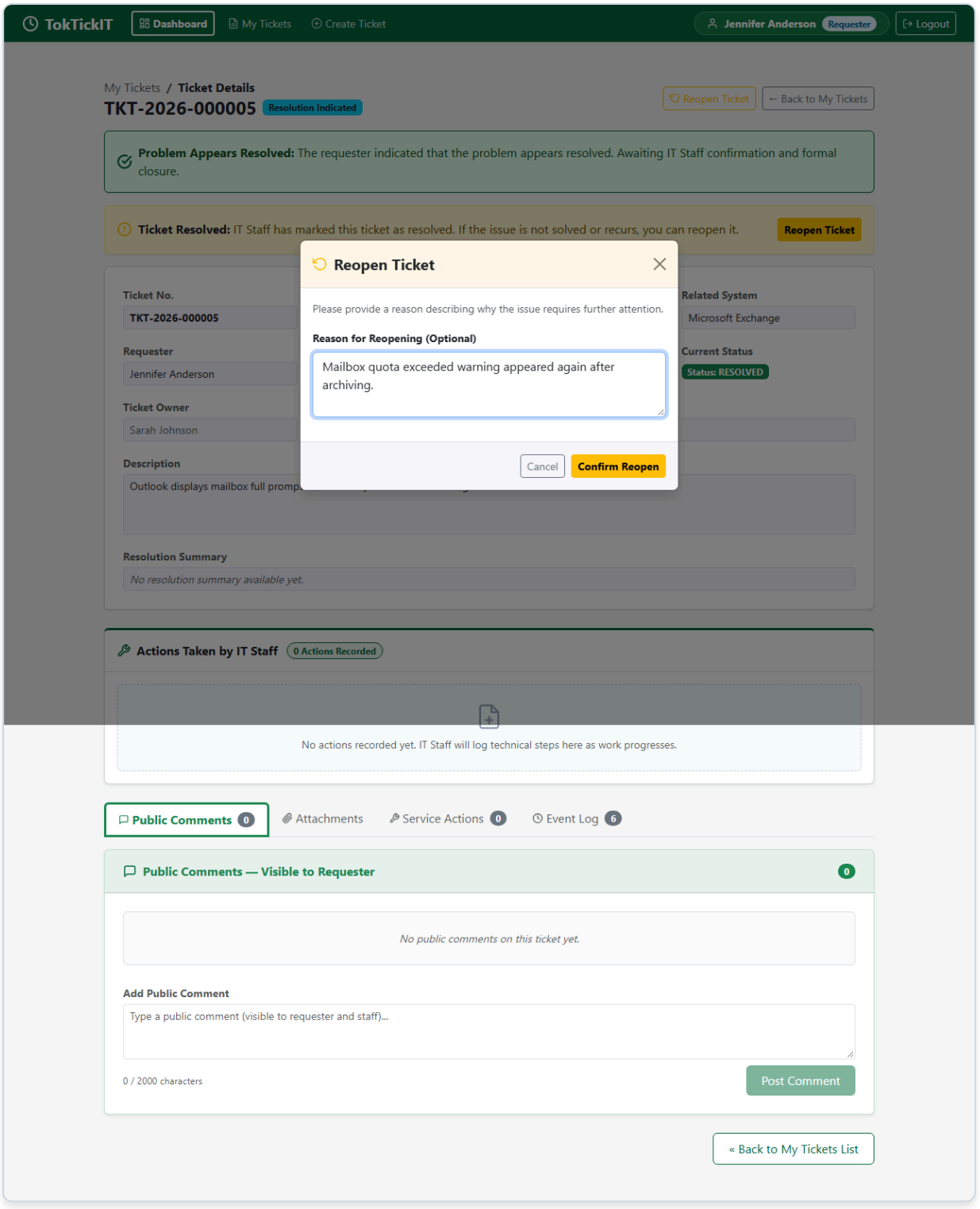
When IT Staff selects `RESOLVED`, an inline `resolution-note-input` textarea appears. Upon saving, the note is automatically posted as a public comment:


TokTickIT
Dashboard
Ticket Queue
Sarah Johnson IT Staff
Logout
Ticket Queue / Ticket Details
TKT-2026-000001 Resolution Indicated
← Back to Ticket Queue
Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.
Ticket No. TKT-2026-000001
Ticket Date Oct 25, 2026, 05:00 PM
Category Network
Related System GlobalProtect VPN
Requester Jennifer Anderson
Requested Priority Priority: HIGH
IT Priority Priority: HIGH
Current Status Status: IN_PROGRESS
Ticket Owner Sarah Johnson
Summary VPN Handshake Gateway Authentication Failure
Description
After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.
Resolution Summary
No resolution summary available yet.
IT Staff Ticket Operations
Staff Control
Change Status
Resolved
Apply
Quick Claim
Claimed by You
Assign Owner
Sarah Johnson (Staff)
IT Priority
HIGH
Resolution Note (Optional)
certificate store flushed on client workstation.
Actions Taken by IT Staff
3 Actions Recorded
+ Add Action Taken
Date / Time
Description
Result / Outcome
Performed By
Assigned To
Status
Follow-Up
Attachm
25 Oct 2026, 18:00
Inspected network gateway RADIUS logs and validated intermediate CA certificate revocation list
Discovered expired intermediate certificate CRL cache on client machine
Sarah Johnson IT Staff
Michael Brown IT Staff
Completed
None
RADI
25 Oct 2026, 21:30
Pushed group policy certificate store refresh and tested client tunnel establishment
Tunnel re-established successfully; monitored connection latency for 30 minutes
Michael Brown IT Staff
Sarah Johnson IT Staff
Completed
Yes
gpo-r
31 Oct 2026, 16:00
Final gateway validation and stability verification under sustained VPN traffic
—
Sarah Johnson IT Staff
Sarah Johnson IT Staff
In Progress
None
—
Public Comments 2
Internal Notes 1
Attachments
Service Actions 3
Event Log 6
Public Comments — Visible to Requester 2
Sarah Johnson IT Staff Oct 25, 2026, 05:30 PM
We have reviewed the network gateway logs. The certificate revocation list (CRL) cache on client side requires a force refresh.
Jennifer Anderson Requester Oct 31, 2026, 09:30 PM
I refreshed the certificate store and connected successfully. ERP access is functioning normally now!
Add Public Comment
Type a public comment (visible to requester and staff)...
0 / 2000 characters
Post Comment
« Back to Ticket Queue

Resolution Note Input

7.4 Requester Reopen Flow & Modal (BR-07 / FR-13 / AC-10)

When a ticket is in `RESOLVED` status, the requester sees a Reopen banner and button. Clicking opens the Reopen Modal to provide a reason and transition status to `REOPENED`:



Requester Reopen Banner and Modal

Answer Part 8: Database & Migration Evidence

8.1 Schema Architecture & `ActionTaken` Prisma Model

The `ActionTaken` model in `server/prisma/schema.prisma` implements clean relational foreign keys:

```
enum ActionTakenStatus {
  PENDING
  IN_PROGRESS
  COMPLETED
  CANCELLED
}

model ActionTaken {
  id                Int              @id @default(autoincrement())
  ticketId          Int
  ticket            Ticket           @relation(fields: [ticketId], references: [id], onDelete: Cascade)
  actionDateTime    DateTime         @default(now())
  actionDescription String           @db.Text
  result            String?          @db.Text
  performedById     Int
  performedBy       User             @relation("ActionPerformer", fields: [performedById], references: [id], onDelete: Restrict)
  assigneeId        Int?
  assignee          User?            @relation("ActionAssignee", fields: [assigneeId], references: [id], onDelete: SetNull)
  status            ActionTakenStatus @default(PENDING)
  followUpRequired  Boolean          @default(false)
  followUpNote      String?          @db.Text
  attachmentNotes   String?          @db.Text
  createdAt         DateTime         @default(now())
  updatedAt         DateTime         @updatedAt

  @@index([ticketId, status])
  @@index([assigneeId])
  @@index([performedById])
}
```

8.2 Down-Migration Strategy & Rollback DDL Execution

In compliance with Handout Section 5.2, the reverse rollback migration DDL was designed and verified via automated test `MIGR-01d`:

```
-- Down-Migration Rollback DDL (docs/lab-04/specification.md Section 7.3)
DROP TABLE IF EXISTS "ActionTaken" CASCADE;
DROP TYPE IF EXISTS "ActionTakenStatus";
```

Automated verification in `server/tests/lab-04/migration-rollback.test.ts` executes this DDL against live foreign-key linked records, proving clean rollback without orphan constraints or database deadlock.

8.3 Idempotent Seeding & Non-Destructive Legacy Support (MIGR-01b)

The seed script (`server/prisma/seed.ts`) idempotently seeds tickets across diverse lifecycle states (0 actions, 1 action, and multiple actions). Legacy tickets with 0 actions render cleanly and function without database errors.

Answer Part 9: Zen Green UI and Responsive Evidence

9.1 Zen Green Color Palette & Semantic Badge System

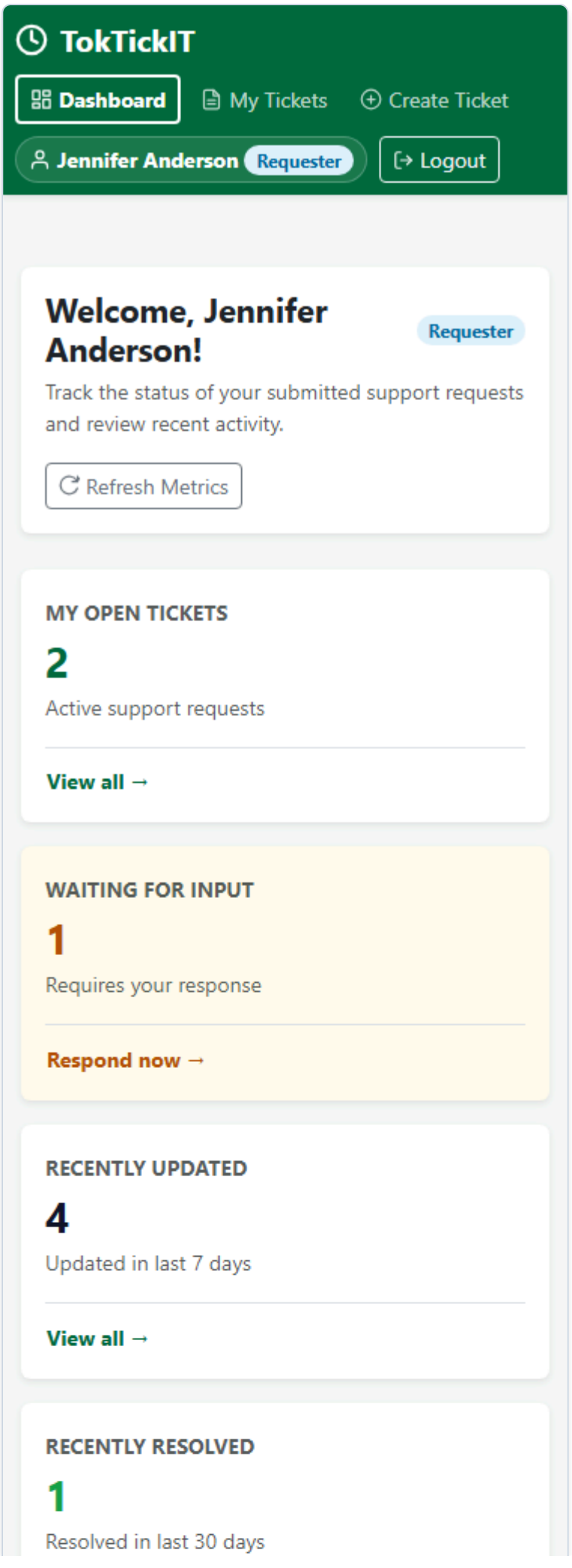
TokTickIT adheres strictly to the Zen Green design system:

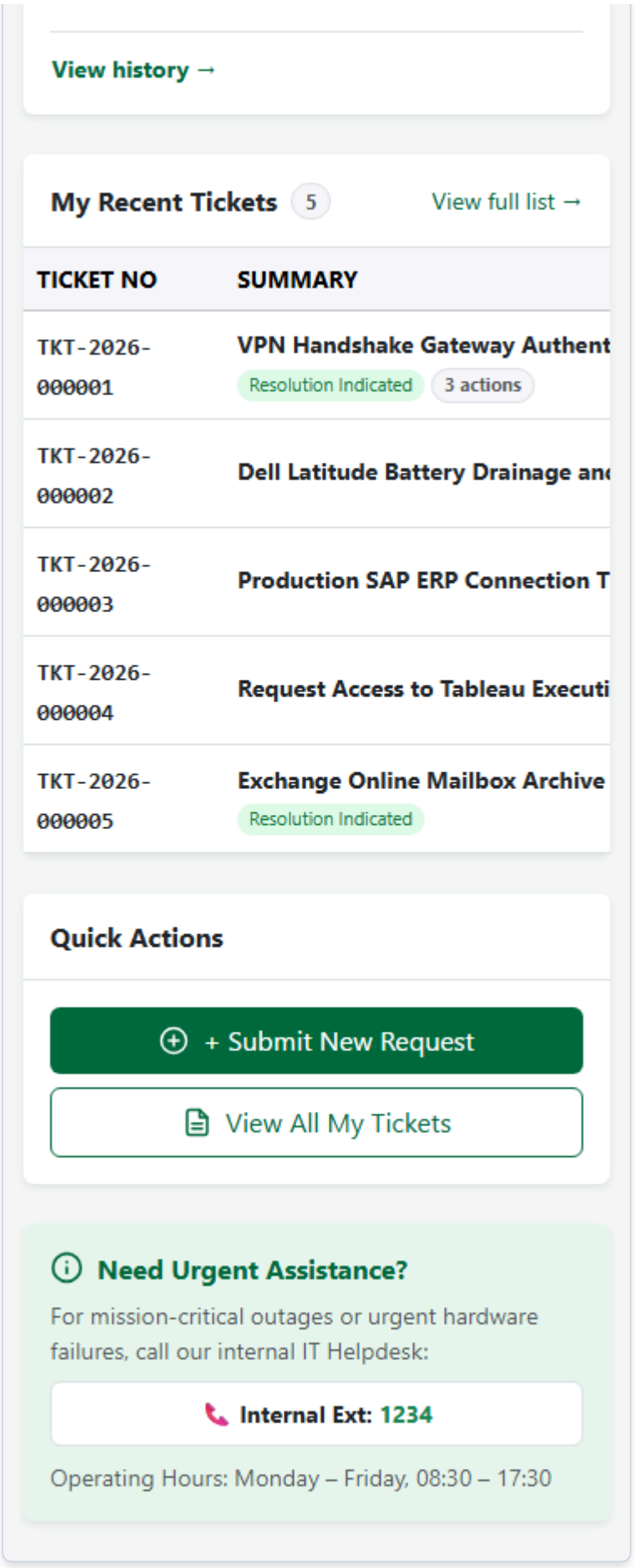
- **Primary Brand**: `#006B3C` (Deep Forest Green)
- **Secondary Brand**: `#0B7A46` (Medium Green)
- **Surface & Backgrounds**: `#F5F7F6` (Page background), `#DCFCE7` (Light green surface)
- **Status Badges**:
  - `NEW`: `#DBEAFE` / `#1E40AF` (Blue)
  - `OPEN`: `#DCFCE7` / `#15803D` (Zen Green)
  - `IN_PROGRESS`: `#FEF3C7` / `#B45309` (Amber)
  - `WAITING_FOR_REQUESTER`: `#FEE2E2` / `#B91C1C` (Coral)
  - `PENDING`: `#E2E8F0` / `#475569` (Slate)
  - `RESOLVED`: `#D1FAE5` / `#065F46` (Mint Green)
  - `REOPENED`: `#FEF08A` / `#854D0E` (Yellow)
  - `CLOSED`: `#1E293B` / `#F8FAFC` (Dark Slate)
  - `CANCELLED`: `#FEE2E2` / `#991B1B` (Red)

9.2 Mobile 375px Zero-Overflow Evidence

All views maintain strict responsive boundaries without horizontal overflow (`overflow-x: hidden`) on 375px mobile viewports:

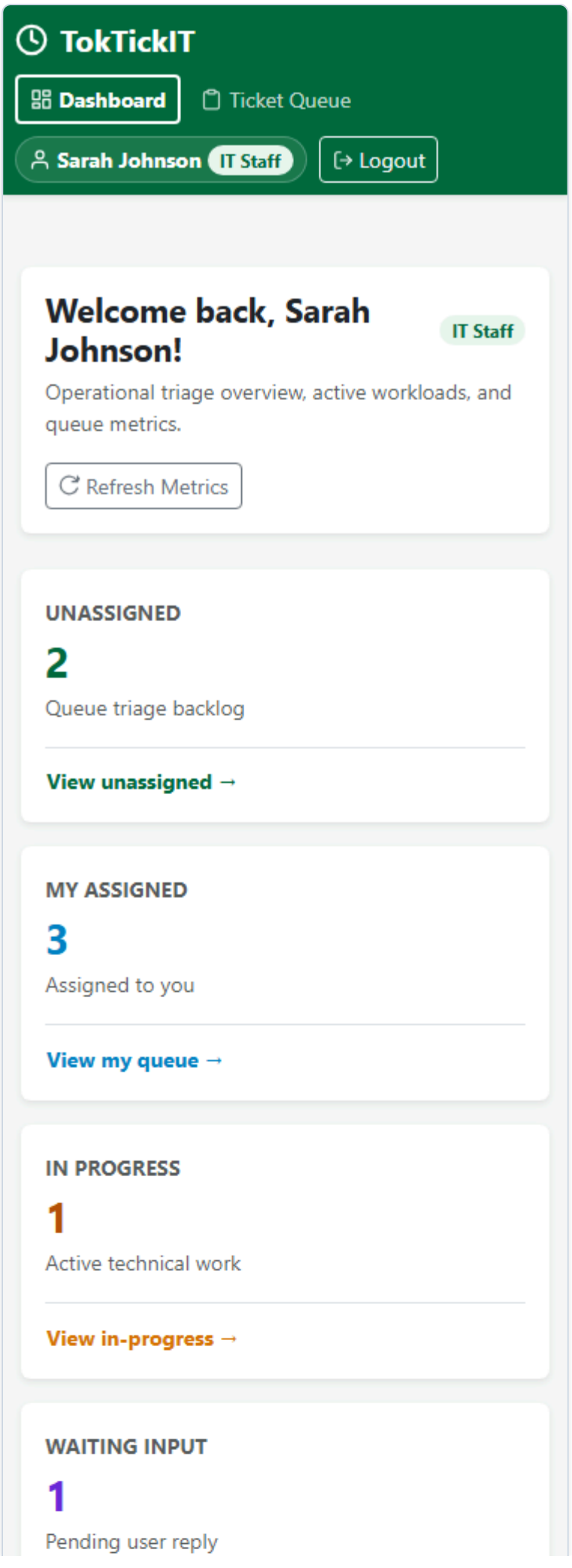
Requester Dashboard at 375px

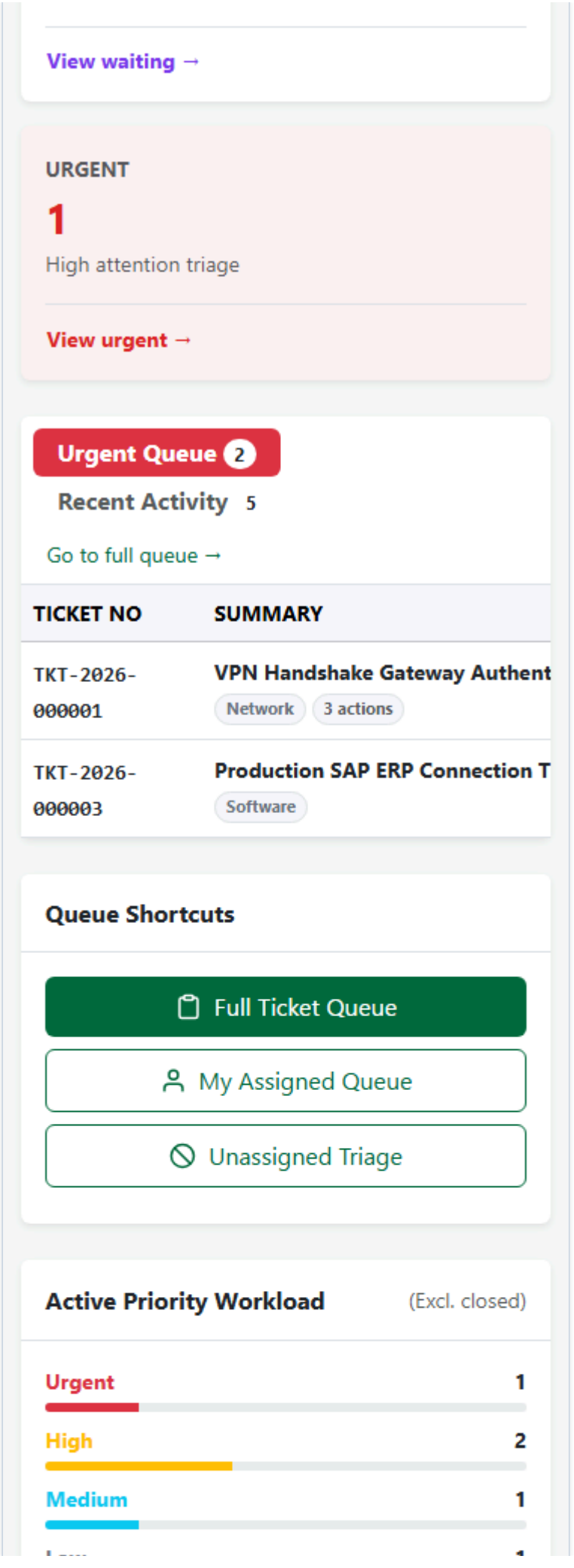

TokTickIT
Dashboard
My Tickets
Create Ticket
Jennifer Anderson
Requester
Logout
Welcome, Jennifer Anderson!
Requester
Track the status of your submitted support requests and review recent activity.
Refresh Metrics
MY OPEN TICKETS
2
Active support requests
View all →
WAITING FOR INPUT
1
Requires your response
Respond now →
RECENTLY UPDATED
4
Updated in last 7 days
View all →
RECENTLY RESOLVED
1
Resolved in last 30 days

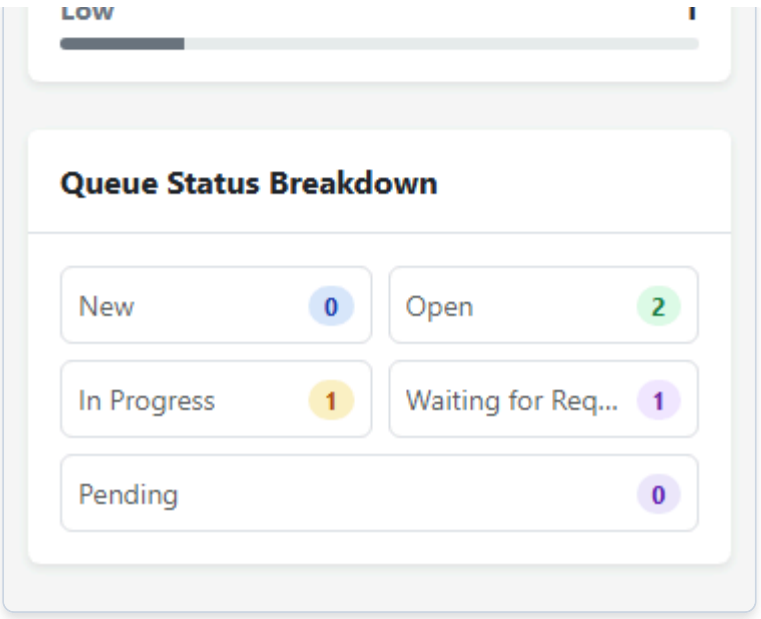


Requester Dashboard 375px

IT Staff Dashboard at 375px

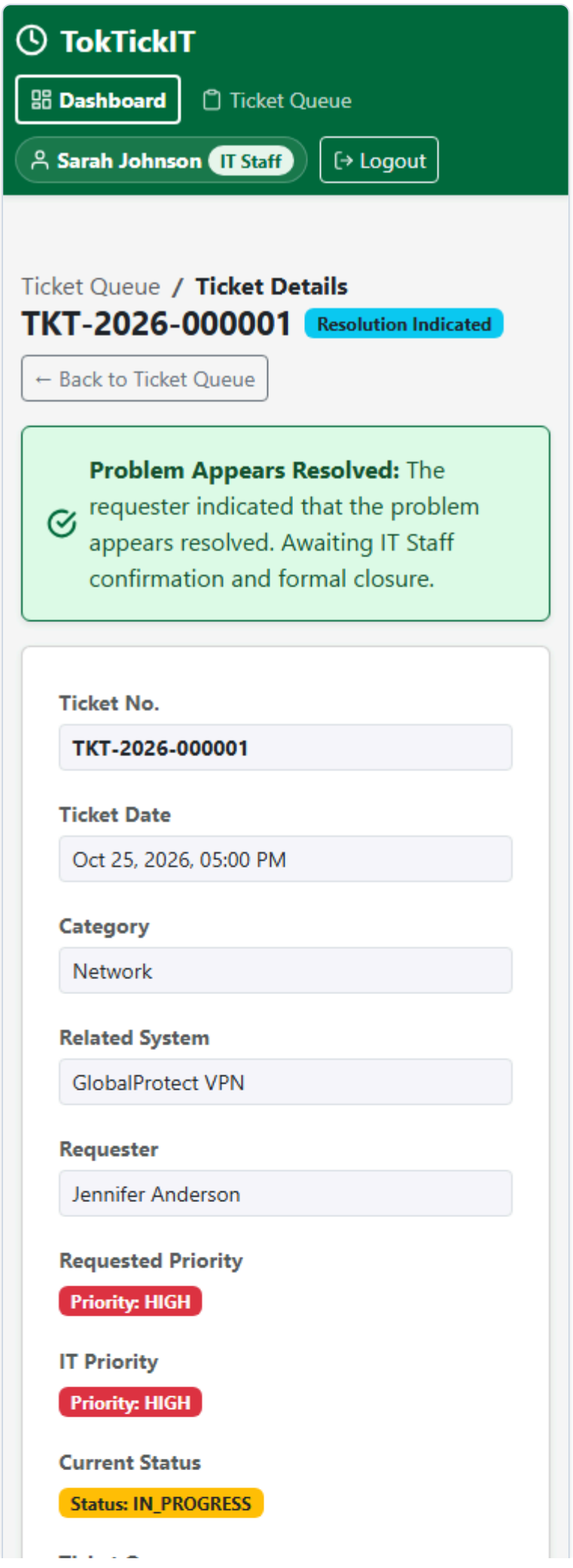

TokTickIT
Dashboard
Ticket Queue
Sarah Johnson
IT Staff
Logout
Welcome back, Sarah Johnson!
IT Staff
Operational triage overview, active workloads, and queue metrics.
Refresh Metrics
UNASSIGNED
2
Queue triage backlog
View unassigned →
MY ASSIGNED
3
Assigned to you
View my queue →
IN PROGRESS
1
Active technical work
View in-progress →
WAITING INPUT
1
Pending user reply


View waiting →
URGENT
1
High attention triage
View urgent →
Urgent Queue 2
Recent Activity 5
Go to full queue →
TICKET NO
SUMMARY
TKT-2026-000001
VPN Handshake Gateway Authent
Network
3 actions
TKT-2026-000003
Production SAP ERP Connection T
Software
Queue Shortcuts
Full Ticket Queue
My Assigned Queue
Unassigned Triage
Active Priority Workload
(Excl. closed)
Urgent 1
High 2
Medium 1



Staff Dashboard 375px

Actions Taken Mobile Cards at 375px


TokTickIT
Dashboard
Ticket Queue
Sarah Johnson
IT Staff
Logout
Ticket Queue / Ticket Details
TKT-2026-000001
Resolution Indicated
← Back to Ticket Queue
Problem Appears Resolved: The requester indicated that the problem appears resolved. Awaiting IT Staff confirmation and formal closure.
Ticket No.
TKT-2026-000001
Ticket Date
Oct 25, 2026, 05:00 PM
Category
Network
Related System
GlobalProtect VPN
Requester
Jennifer Anderson
Requested Priority
Priority: HIGH
IT Priority
Priority: HIGH
Current Status
Status: IN_PROGRESS

Ticket Owner

Sarah Johnson

Summary

VPN Handshake Gateway Authentication Failure

Description

After the corporate security policy update, IPSec VPN fails during handshake with error code SEC_ERROR_UNTRUSTED_ISSUER. Requester cannot access financial reporting tools.

Resolution Summary

No resolution summary available yet.

IT Staff Ticket Operations Staff Control

Quick Claim

Claimed by You

Assign Owner

Sarah Johnson (Staff)

IT Priority

HIGH

Change Status

Select Next Status... Apply

Actions Taken by IT Staff 3 Actions Recorded

+ Add Action Taken

25 Oct 2026, 18:00 Completed

Inspected network gateway RADIUS logs and validated intermediate CA

certificate revocation list

Outcome: Discovered expired intermediate certificate CRL cache on client machine

By: Sarah Johnson •

Assignee: Michael Brown

RADIUS-log-dump-20261025.txt

25 Oct 2026, 21:30 Completed

Pushed group policy certificate store refresh and tested client tunnel establishment

Outcome: Tunnel re-established successfully; monitored connection latency for 30 minutes

By: Michael Brown •

Assignee: Sarah Johnson

Follow-up: Verify certificate auto-enrollment persists through next scheduled domain sync

gpo-result-report.pdf

31 Oct 2026, 16:00 In Progress

Final gateway validation and stability verification under sustained VPN traffic

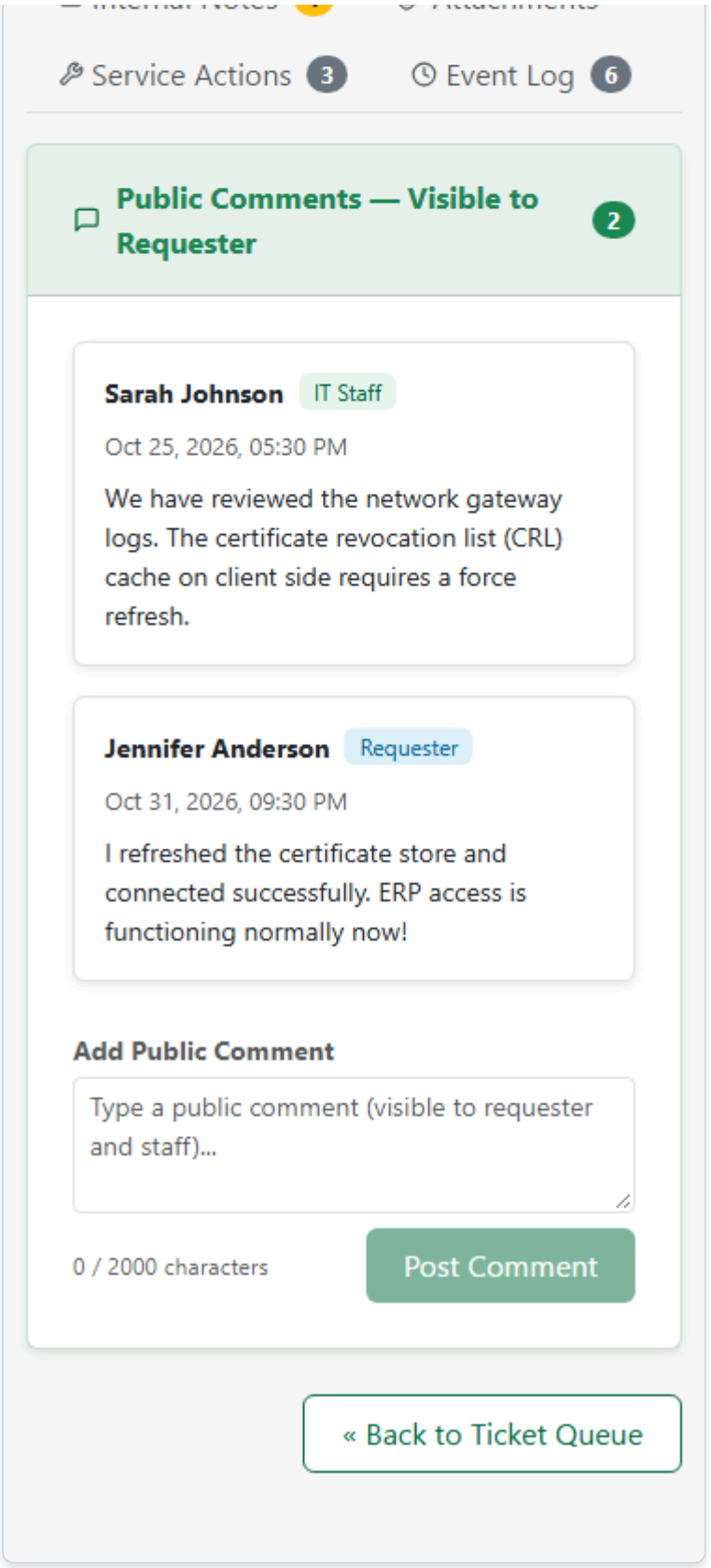
By: Sarah Johnson •

Assignee: Sarah Johnson

Edit Action Status ▾

Public Comments 2

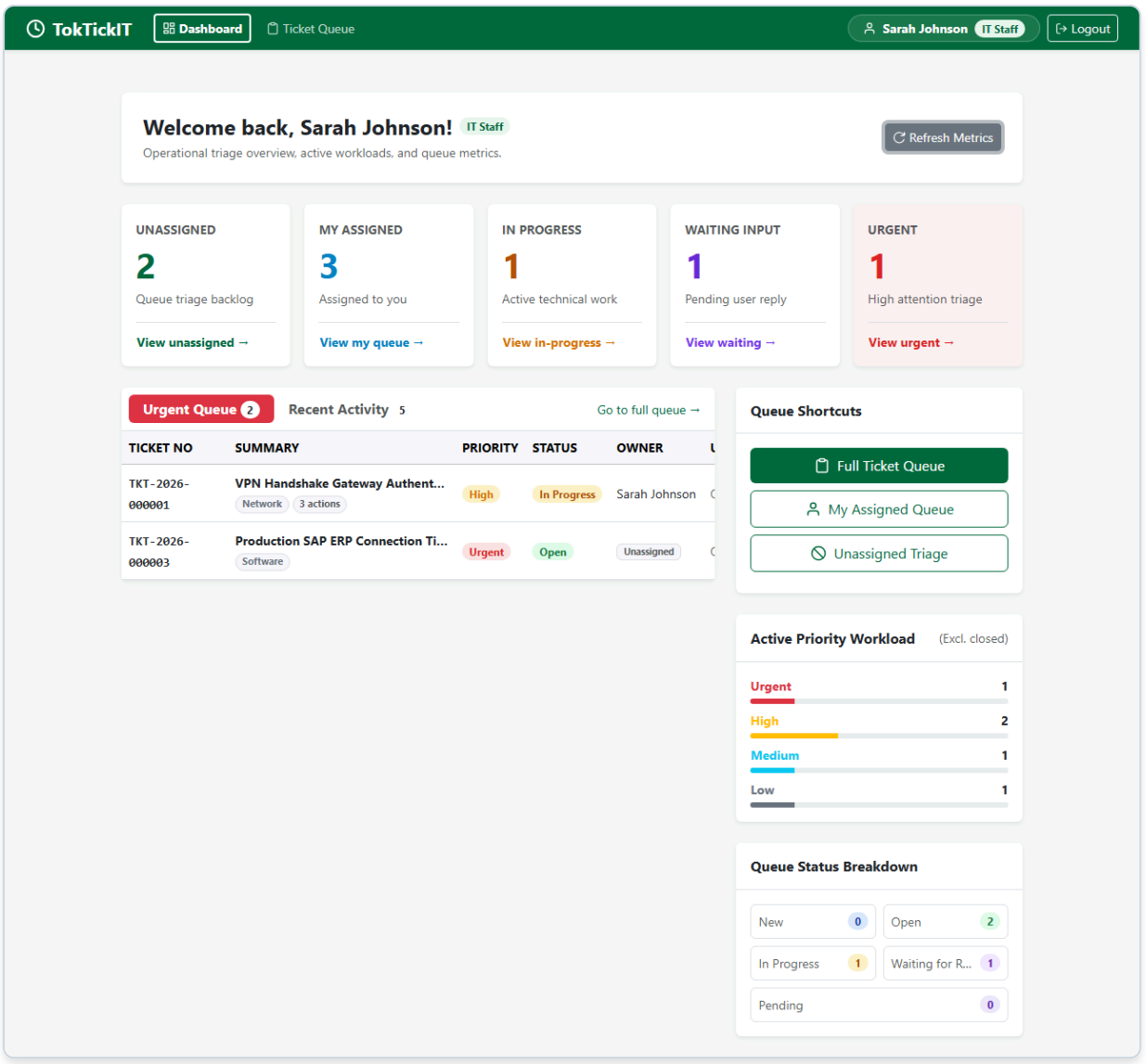
Internal Notes 1 Attachments



Actions Taken 375px

9.3 Keyboard Focus Ring & Accessibility Verification (A11Y-01)

All interactive controls display prominent, high-contrast focus rings when navigated via keyboard Tab key, adhering to WCAG 2.1 AA accessibility guidelines:



Keyboard Focus Ring

Conclusion & Verification Sign-Off

TokTickIT Lab 4 delivers all functional requirements, business rules, role dashboards, Actions Taken subsystems, resolution gates, and final regression tests with 100% test pass rate (375/375 automated tests) and zero compilation warnings.